

The `coppe` document class

Version 3.8

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typeset in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will be used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@teseph`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (`brazilian` by default; one of `english`, `spanish`, `french`, `italian` also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for brazilian/english in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys available from, urlseen, in, editor, editors, depositor, coppescale, msdiss, dscthis, phdthesis, tcc. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}           % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\~erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol.      \end{abstract}
  \begin{foreignabstract} English foreign abstract.          \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\~on} Cuerpo en espa\~nol\dots
\end{document}
```


6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
brazilian (default)	— (built into <code>coppe.cls</code>)
english	— (built into <code>coppe.cls</code>)
spanish	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
french	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
italian	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Quick, useful tips

Pictures. The default picture format of $\text{\LaTeX}$ is the Encapsulated PostScript (EPS). If you use `pdf\text{\LaTeX}`, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdf\text{\LaTeX}` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in $\text{\LaTeX}$ is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under `pdf\text{\LaTeX}`: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (Lua $\text{\LaTeX}$, Xe $\text{\LaTeX}$). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math`

nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

8 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating} % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz} % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23
24 \begin{document}
25 \title{Titulo da Tese}
26 \foreigntitle{Thesis Title}
27 \author{Nome do Autor}{Sobrenome}
28 \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
```

```

29 \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
30 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
31 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
32 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
33 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
34
35 \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
36 \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
37 \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
38 \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
39 \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
40 \department{PESC}
41 \date{01}{2024}
42
43 \keyword{Primeira palavra-chave}
44 \keyword{Segunda palavra-chave}
45 \keyword{Terceira palavra-chave}
46
47 \maketitle
48
49 \frontmatter
50 \dedication{A alguém cujo valor é digno desta dedicatória.}
51
52 \chapter*{Agradecimentos}
53
54 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
55 orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
56 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
57 ut labore et dolore magna aliqua.
58
59 \begin{abstract}
60
61 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
62 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
63 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
64 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
65 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
66 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
67 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
68 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
69 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
70 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
71 aspernatur aut odit aut fugit.
72
73 \end{abstract}
74
75 \begin{foreignabstract}
76
77 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
78 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
79 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
80 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
81 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
82 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui

```

```

83 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
84 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
85 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
86 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
87 aspernatur aut odit aut fugit.
88
89 \end{foreignabstract}
90
91 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
92 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
93 \listoffigures
94 \listoftables
95 \listofquadros
96 \listofprogramas
97 \listofalgorithms
98 \printloabbreviations
99 \printlosymbols
100 \tableofcontents
101
102 \mainmatter
103 \chapter{Introdução}
104
105 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
106
107 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
108
109 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
110
111 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
112
113 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
114
115 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
116
117 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
118
119 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
120
121 Este documento não substitui, mas complementa, o documento que descreve a classe.
122
123
124
125 \chapter{Configurações Iniciais}
126
127 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
128
129 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
130
131 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
132
133 \section{Linguagem principal do texto}
134
135 Essa classe considera que o texto principal está em português e algumas partes específicas
136 \begin{itemize}

```

137 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.

138 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

139 \end{itemize}

140

141 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CopeT

142

143

144 \section{Por que usar o \LaTeX}

145

146 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess

147

148 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p

149

150 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

151

152 \section{Como e onde usar o \LaTeX}

153

154 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente

155 \begin{enumerate}

156 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win

157 \item Usar um ambiente na rede, como o Overleaf.

158 \end{enumerate}

159

160 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b

161

162

163 \chapter{Algumas Regras da COPPE}

164

165 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

166

167 É imprescindível definir os símbolos, tal como o

168 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.

169 \syml{\mathbb{R}}{Conjunto dos números reais}

170 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como

171

172 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve

173

174 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice

175

176 \section{Citações}

177 \label{sec:citacoes}

178

179 Citações curtas podem ser feitas \quote{o comando quote} ou direto com “duas crases e dois

180

181 \begin{longquote}

182 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)

183 feita com o ambiente \verb=longquote= The primary objective of this

184 investigation was to determine the feasibility of detecting corrosion in

185 aluminum Naval aircraft components with neutron radiographic interrogation

186 and the use of standard corrosion penetrameters. Secondary objectives

187 included the determination of the effect of object thickness on image quality,

188 the defining of minimum levels of detectability and a preliminary investigation

189 of a means whereby the degree of corrosion could be quantified with neutron

190 radiographic data. \cite{article-example}

```

191 \end{longquote}
192
193 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
194
195 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
196
197 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref
198
199 \chapter{Floats}
200
201 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
202
203 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
204
205 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
206
207 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
208
209
210 \section{Tabelas e Figuras Padrão}
211
212 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
213
214 \begin{table}[ht]
215 \centering % Centraliza a tabela
216 \caption{Exemplo de Tabela de Números}
217 \label{tab:exemplo_numeros}
218 \begin{tabular}{ccc} % Define a quantidade de colunas
219 \hline % Linha superior
220 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
221 \hline % Linha média
222 1 & 2 & 3 \\ % Primeira linha de dados
223 \hline
224 4 & 5 & 6 \\ % Segunda linha de dados
225 \hline
226 7 & 8 & 9 \\ % Terceira linha de dados
227 \hline
228 10 & 11 & 12 \\ % Quarta linha de dados
229 \hline % Linha inferior
230 \end{tabular}
231 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatória, abaixo do float
232 \end{table}
233
234
235
236 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
237
238 \begin{figure}[ht]
239 \centering % Centraliza a figura
240 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
241 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
242 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
243 \source{COPPE/UFRJ.} % fonte obrigatória, abaixo da figura
244 \end{figure}

```

```

245
246 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
247
248 \begin{quadro}[ht]
249 \centering
250 \caption{Exemplo de Quadro}
251 \label{qua:exemplo}
252 \begin{tabular}{|l|l|}
253 \hline
254 \textbf{Ilustração} & \textbf{Delimitação} \\
255 \hline
256 Tabela & Linhas apenas no topo e na base. \\
257 \hline
258 Quadro & Linhas em todas as bordas. \\
259 \hline
260 \end{tabular}
261 \fonte{Elabora\c{c}\~ao pr\'opria.}
262 \end{quadro}
263
264
265
266
267
268 \section{Tabelas mais elegantes}
269
270 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
271
272 \begin{table}[ht]
273 \centering % Centraliza a tabela
274 \caption{Exemplo de Tabela de Números mais elegantes}
275 \label{tab:exemplo_numerosbom}
276 \begin{tabular}{ccc} % Define a quantidade de colunas
277 \toprule % Linha superior
278 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
279 \midrule % Linha média
280 1 & 2 & 3 \\
281 4 & 5 & 6 \\
282 7 & 8 & 9 \\
283 10 & 11 & 12 \\
284 \bottomrule % Linha inferior
285 \end{tabular}
286 \fonte{Elabora\c{c}\~ao pr\'opria.}
287 \end{table}
288
289 \section{Tabelas Longas ou Largas}
290
291 Se sua tabela é muito longa ou larga, existem várias opções.
292 \begin{itemize}
293 \item alterar o tamanho da letra
294 \item Usar o longtable
295 \item rodar a tabela, fazendo ela em \textit{landscape}
296 \item fazer a tabela dentro de um minibox
297 \end{itemize}
298

```



```

299
300 \subsection{Tabelas largas demais}
301
302 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
303
304 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
305
306 \begin{table}[ht]
307 \centering % Centraliza a tabela
308 \caption{Exemplo de Tabela Larga com Fonte Menor}
309 \label{tab:tabela_largafns}
310 \footnotesize % Aplica uma fonte menor para a tabela
311 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
312 \toprule
313 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} \\
314 \midrule
315 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
316 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
317 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
318 \bottomrule
319 \end{tabular}
320 \fonte{Elabora\c{c}\~ao pr\'opria.}
321 \end{table}
322
323 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
324
325 \begin{table}[ht]
326 \centering
327 \caption{Exemplo de Tabela Redimensionada}
328 \label{tab:examplerb}
329 \resizebox{\textwidth}{!}{%
330 \begin{tabular}{llll}
331 \toprule
332 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
333 \midrule
334 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
335 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
336 \bottomrule
337 \end{tabular}%
338 }
339 \fonte{Elabora\c{c}\~ao pr\'opria.}
340 \end{table}
341
342
343 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
344
345 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
346
347 \begin{sidewaystable}
348 \centering
349 \caption{Sua Legenda Aqui}
350 \label{tab:sua_tabela}
351 \begin{tabular}{llll}
352 \toprule

```

```

353 Coluna 1 & Coluna 2 & Coluna 3 \\
354 \midrule
355 Item 1 & Item 2 & Item 3 \\
356 Item 4 & Item 5 & Item 6 \\
357 \bottomrule
358 \end{tabular}
359 \fonte{Elabora\c{c}\~ao pr\'opria.}
360 \end{sidewaystable}
361
362 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
363
364 % Exemplo de tabela longa que se estende por várias páginas
365 \begin{longtable}{|c|c|c|}
366 % primeiro cabeçalho (é o caption)
367 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
368 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
369 \endfirsthead
370 % cabeçalho normal
371 \multicolumn{3}{c}{%
372 {\table\thetable} -- continuação da página anterior}} \\
373 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
374 \endhead
375 % pé normal
376 \hline \multicolumn{3}{|r|}{Continua na próxima página} \\ \hline
377 \endfoot
378 \hline
379 % último pé
380 \multicolumn{3}{|r|}{Continua na próxima página}} \\
381 \hline \hline
382 \endlastfoot
383
384 % Conteúdo da tabela
385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
395 1 & 2 & 3 \\
396 4 & 5 & 6 \\
397 1 & 2 & 3 \\
398 4 & 5 & 6 \\
399 1 & 2 & 3 \\
400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 1 & 2 & 3 \\
403 4 & 5 & 6 \\
404 1 & 2 & 3 \\
405 4 & 5 & 6 \\
406 1 & 2 & 3

```

407 4 & 5 & 6 \\
 408 1 & 2 & 3 \\
 409 4 & 5 & 6 \\
 410 1 & 2 & 3 \\
 411 4 & 5 & 6 \\
 412 1 & 2 & 3 \\
 413 4 & 5 & 6 \\
 414 1 & 2 & 3 \\
 415 4 & 5 & 6 \\
 416 1 & 2 & 3 \\
 417 4 & 5 & 6 \\
 418 1 & 2 & 3 \\
 419 4 & 5 & 6 \\
 420 1 & 2 & 3 \\
 421 4 & 5 & 6 \\
 422 1 & 2 & 3 \\
 423 4 & 5 & 6 \\
 424 1 & 2 & 3 \\
 425 4 & 5 & 6 \\1 & 2 & 3 \\
 426 4 & 5 & 6 \\
 427 1 & 2 & 3 \\
 428 4 & 5 & 6 \\
 429 1 & 2 & 3 \\
 430 1 & 2 & 3 \\
 431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 4 & 5 & 6 \\
 436 1 & 2 & 3 \\
 437 4 & 5 & 6 \\
 438 1 & 2 & 3 \\
 439 4 & 5 & 6 \\
 440 1 & 2 & 3 \\
 441 4 & 5 & 6 \\
 442 1 & 2 & 3 \\
 443 4 & 5 & 6 \\
 444 1 & 2 & 3 \\
 445 4 & 5 & 6 \\
 446 1 & 2 & 3 \\
 447 4 & 5 & 6 \\
 448 1 & 2 & 3 \\
 449 4 & 5 & 6 \\
 450 1 & 2 & 3 \\
 451 4 & 5 & 6 \\
 452 1 & 2 & 3 \\
 453 4 & 5 & 6 \\
 454 1 & 2 & 3 \\
 455 4 & 5 & 6 \\
 456 1 & 2 & 3 \\
 457 4 & 5 & 6 \\
 458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
 460 1 & 2 & 3 \\

```

461 4 & 5 & 6 \\
462 1 & 2 & 3 \\
463 4 & 5 & 6 \\
464 1 & 2 & 3 \\
465 4 & 5 & 6 \\
466 1 & 2 & 3 \\
467 4 & 5 & 6 \\
468 1 & 2 & 3 \\
469 4 & 5 & 6 \\
470 1 & 2 & 3 \\
471 4 & 5 & 6 \\
472 1 & 2 & 3 \\
473 4 & 5 & 6 \\
474 1 & 2 & 3 \\
475 4 & 5 & 6 \\
476 1 & 2 & 3 \\
477 4 & 5 & 6 \\
478 1 & 2 & 3 \\
479 4 & 5 & 6 \\
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486 1 & 2 & 3 \\
487 4 & 5 & 6 \\
488 1 & 2 & 3 \\
489 4 & 5 & 6 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492
493 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
494 \end{longtable}
495 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
496
497 \section{Subfiguras (subcaption)}
498
499 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
500
501 \begin{figure}[ht]
502   \centering
503   \begin{subfigure}[b]{0.42\textwidth}
504     \centering\includegraphics[width=\linewidth]{example-image-a}
505     \caption{Primeira vista}\label{fig:sub-a}
506   \end{subfigure}
507   \hfill
508   \begin{subfigure}[b]{0.42\textwidth}
509     \centering\includegraphics[width=\linewidth]{example-image-b}
510     \caption{Segunda vista}\label{fig:sub-b}
511   \end{subfigure}
512   \caption{Figura com duas subfiguras}
513   \label{fig:subs}
514   \fonte{Elabora\c{c}\~ao pr\'opria.}

```

```

515 \end{figure}
516
517 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fi
518
519 \section{Imagens (graphicx)}
520
521 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
522 \begin{itemize}
523   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
524   \item \verb|[angle=90]| --- rotação;
525   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, dire
526 \end{itemize}
527 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt
528
529 \section{Introdução ao tikz}
530
531 O \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
532
533 \begin{figure}[ht]
534   \centering
535   \begin{tikzpicture}
536     \node[draw, rounded corners] (a) at (0,0) {Início};
537     \node[draw, rounded corners] (b) at (4,0) {Fim};
538     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
539   \end{tikzpicture}
540   \caption{Diagrama simples em tikz}
541   \label{fig:tikz}
542   \fonte{Elabora\c{c}\~ao pr\'opria.}
543 \end{figure}
544
545 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
546
547 \section{Float ou direto no texto?}
548
549 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
550
551 \chapter{Using BibLaTeX}
552 \label{cap:biblatex}
553
554 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
555
556 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequênci
557
558 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
559
560 \begin{table}[h]
561 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
562   \texttt{\textbackslash cite} do \LaTeX\ e
563   o comando \texttt{\textbackslash citet},
564   disponível pela opção \texttt{natbib} do BibLaTeX.}
565 \label{tab:citation}
566 \centering
567 {\footnotesize
568 \begin{tabular}{|c|c|c|}

```

```

569 \hline
570 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
571 \hline
572 Livro & \cite{book-example} & \citet{book-example}\
573 Artigo & \cite{article-example} & \citet{article-example}\
574 Relatório & \cite{techreport-example} & \citet{techreport-example}\
575 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
576 Anais de Congresso & \cite{inproceedings-example} &
577 \citet{inproceedings-example}\
578 Séries & \cite{incollection-example} & \citet{incollection-example}\
579 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
580 Dissertação de mestrado & \cite{mastersthesis-example} &
581 \citet{mastersthesis-example}\
582 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
583 \hline
584 \end{tabular}}
585 \fonte{Elabora\c{c}\~ao pr\'opria.}
586 \end{table}
587
588 \begin{table}[h]
589 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
590 \texttt{\textbackslash cite} do \LaTeX\ e
591 o comando \texttt{\textbackslash citet},
592 disponível pela opção \texttt{\textbackslash natbib} do BibLaTeX. Além disso, usando o booktabs.}
593 \label{tab:citation1}
594 \centering
595 {\footnotesize
596 \begin{tabular}{ccc}
597 \toprule
598 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
599 \midrule
600 Livro & \cite{book-example} & \citet{book-example}\
601 Artigo & \cite{article-example} & \citet{article-example}\
602 Relatório & \cite{techreport-example} & \citet{techreport-example}\
603 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
604 Anais de Congresso & \cite{inproceedings-example} &
605 \citet{inproceedings-example}\
606 Séries & \cite{incollection-example} & \citet{incollection-example}\
607 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
608 Dissertação de mestrado & \cite{mastersthesis-example} &
609 \citet{mastersthesis-example}\
610 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
611 \bottomrule
612 \end{tabular}}
613 \fonte{Elabora\c{c}\~ao pr\'opria.}
614 \end{table}
615
616 \section{Comandos de citação}
617
618 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
619
620 \subsection{Parâmetro opcional: página e “tradução nossa”}
621
622 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser

```

```

623
624 \subsection{Citação de citação (\emph{apud})}
625
626 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
627
628 \subsection{Múltiplas bibliografias}
629
630 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
631
632 \section{Tipos de entrada bibliográfica}
633
634 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
635
636 \subsection{Tipos acadêmicos principais}
637 \begin{itemize}
638 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
639 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
640 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
641 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
642 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
643 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); períod
644 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
645 \end{itemize}
646
647 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
648 \begin{itemize}
649 \item Patente --- \verb|@patent| (\verb|@patente|)
650 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
651 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\v
652 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
653 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\ver
654 \end{itemize}
655
656 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
657 \begin{itemize}
658 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
659 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
660 \end{itemize}
661
662 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
663
664 \chapter{Alguns outros exemplo úteis}
665
666 \begin{tcolorbox}[title=Meu Textbox]
667 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
668 \end{tcolorbox}
669
670 \begin{tcolorbox}
671 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
672 \end{tcolorbox}
673
674 \begin{figure}[ht]
675 \centering
676 \begin{tikzpicture}

```

```

677 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
678 \begin{scope}[x={(image.south east)},y={(image.north west)}]
679 % Definindo o textbox dentro da figura
680 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
681 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
682 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
683 \end{tcolorbox}
684 ];
685 \end{scope}
686 \end{tikzpicture}
687 \caption{Figura com Textbox}
688 \label{fig:figura_com_textbox1}
689 \fonte{Elabora\c{c}\~ao pr\'opria.}
690 \end{figure}
691
692
693 \begin{figure}[ht]
694 \centering
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
697 \end{tcolorbox}
698 \caption{Figura com Textbox simples}
699 \label{fig:figura_com_textbox}
700 \fonte{Elabora\c{c}\~ao pr\'opria.}
701 \end{figure}
702
703 \section{Listagens de código e algoritmos}
704
705 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
706
707 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
708 def saudacao(nome):
709     # imprime uma saudação acentuada
710     print(f"Olá, {nome}!")
711
712 saudacao("COPPE")
713 \end{python}
714 \source{Elaboração própria.}
715
716 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
717
718 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
719 Olá, COPPE!
720 \end{Verbatim}
721
722 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
723
724 \begin{algorithm}[H]
725 \caption{Soma dos elementos de um vetor}
726 \label{alg:soma}
727 \Dados{vetor $v$ com $n$ elementos}
728 \Resultado{soma $s$ dos elementos de $v$}
729 $s \leftarrow 0$;
730 \Para{$i \leftarrow 1$ \Ate{} $n$}{

```



```

731  $s \leftarrow s + v[i]$\;
732 }
733 \Retorna $s$\;
734 \end{algorithm}
735
736 \chapter{Escrevendo matemática}
737
738 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carrega
739
740 \section{Alinhamento (align)}
741 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebrando a linha)
742 \begin{align}
743   f(x) &= (x+1)^2 \\
744         &= x^2 + 2x + 1 .
745 \end{align}
746
747 \section{Matrizes (matrix)}
748 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
749 \begin{equation}
750   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
751   \quad \quad \quad \\
752   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\
753 \end{equation}
754
755 \section{Subequações (subequations)}
756 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
757 \begin{subequations}
758 \begin{align}
759   a &= b + c , \\
760   d &= e - f .
761 \end{align}
762 \end{subequations}
763
764 \section{Estruturas alinhadas (array)}
765 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
766 \begin{equation}
767   |x| = \left\{ \begin{array}{l} x, \quad \text{se } x \geq 0 , \\ -x, \quad \text{caso contrário.} \end{array} \right. \\
770 \end{equation}
771
772
773 \section{Operadores (\textbackslash DeclareMathOperator)}
774 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no preâmbulo:
775 \begin{verbatim}
776 \usepackage{amsmath}
777 \DeclareMathOperator{\RMSE}{RMSE}
778 \DeclareMathOperator{\diag}{diag}
779 \end{verbatim}
780 Assim, \verb|$ \RMSE(y, \hat{y})$| produz  $\text{RMSE}(y, \hat{y})$ , e \verb|$ \diag(\lambda_1, \dots, \lambda_n)$| produz  $\text{diag}(\lambda_1, \dots, \lambda_n)$ 
781
782 \section{Unicode e LuaLaTeX (unicode-math)}
783 \label{sec:unicode-math}
784 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amsmath}

```

```

785 \begin{verbatim}
786 \usepackage{amsmath}      % a classe já carrega
787 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
788 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
789 \end{verbatim}
790 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
791
792 \chapter{Referências cruzadas, notas e índices}
793 \label{cap:refs}
794
795 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
796
797 \section{Rótulos e referências cruzadas}
798
799 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
800
801 \section{Notas de rodapé}
802
803 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s
804
805 \section{hyperref e marcadores}
806
807 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
808
809 \section{URLs}
810
811 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
812
813 \section{Índice remissivo}
814
815 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
816
817 \section{Traduções (NBR 10520)}
818
819 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
820
821 \chapter{Truques de \LaTeX}
822
823 \section{Comentários mágicos (\texttt{\% !TeX})}
824 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
825 \begin{verbatim}
826 % !TeX program = lualatex
827 % !TeX encoding = UTF-8
828 % !TeX spellcheck = pt_BR
829 % !BIB program = biber
830 \end{verbatim}
831 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
832
833 \section{Espaço engolido}
834 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
835
836 \section{Escopo (grupos)}
837 Chaves \verb|\{...}| (ou \verb|\begingroup\...\endgroup|) criam um escopo: mudanças
838

```

```

839 \section{Descarregar figuras: \texttt{\textbackslash clearpage} * \texttt{\textbackslash m
840 \verb|newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
841
842 \section{\texttt{latexmkrc}, perl e Overleaf}
843 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
844
845 \chapter{Método Proposto}
846 \chapter{Resultados e Discussão}
847
848 \section{Algumas Demonstrações}
849
850 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
851 \syml[beta]{Beta}{A palavra Beta mais e corrigida}
852 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
853 \syml[beta]{A palavra beta}
854 \syml[alpha]{A palavra alpha}
855 \syml[alpha]{Alpha}{A palavra Alpha}
856 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
857 \syml[marco]{Marco}{A palavra Marco corrigida}
858
859 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
860 \abbrev{GOT}{Game of Thrones}
861 \abbrev[GOT]{GOT}{Game of Thrones ordenado como GOT}
862 \abbrev[iot]{IoT}{IoT ordenado como iot}
863 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
864 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
865 \abbrev{IoT}{IoT com ordenação default}
866 \abbrev[ITU]{ITU}{ITU mesmo}
867
868 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
869 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
870
871 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
872
873
874
875
876 \chapter{Conclusões}
877
878 \backmatter
879
880 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
881 % partitura --- declarados via sinônimos em português em example.bib.
882 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
883 \nocite{jurisprudencia-example}
884 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
885 \nocite{jornalcaderno-example,jornal-example}
886 \printbibliography
887
888
889
890 \appendix
891
892 \chapter{Um apêndice}

```

```

893
894 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
895
896 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
897
898
899 \chapter{Outro apêndice}
900
901 \annex
902
903
904 \chapter{Um Anexo}
905 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
906
907
908
909 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
910
911 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comando
912
913
914 \chapter{Outro Anexo}
915
916
917 \end{document}
918
919 \example

```

9 Implementation

9.1 The ‘coppe.cls’ file

```

920 \class
921 \def\filename{coppe.dtx}
922 \def\fileversion{v4.0}
923 \def\filedate{2026/05/28}
924 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
925 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
926 \LoadClass[12pt,a4paper,twoside]{book}
927 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
928 % the END of the class (after every class hook is defined), in that order:
929 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
930 % it patches (TOC, refs, lists) are already defined, and biblatex must come
931 % after hyperref so its citation links and back-references work.
932 \RequirePackage{hyphenat}
933 \RequirePackage{lastpage}
934 \RequirePackage{ifthen}
935 \RequirePackage{graphicx}
936 \RequirePackage{tabularx}
937 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
938 \RequirePackage{longtable} % tabelas que quebram entre páginas
939 \RequirePackage{etoolbox}
940 \RequirePackage{ltxcmds}
941 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...

```

```

942 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
943 \def\TeX{\hologo{TeX}}
944 \def\LaTeX{\hologo{LaTeX}}
945 \def\LaTeXe{\hologo{LaTeXe}}
946 \def\BibTeX{\hologo{BibTeX}}
947 \def\pdfLaTeX{\hologo{pdfLaTeX}}
948 \def\pdfTeX{\hologo{pdfTeX}}
949 \def\LuaLaTeX{\hologo{LuaLaTeX}}
950 \def\XeLaTeX{\hologo{XeLaTeX}}
951 \def\MiKTeX{\hologo{MiKTeX}}
952 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc; LuaLaTeX (and XeTeX)
953 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
954 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
955 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
956 \RequirePackage{iftex}
957 \ifPDFTeX
958   \RequirePackage[utf8]{inputenc}
959   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
960     Although slower, you should try compiling with LuaLaTeX}}
961 \fi
962 \RequirePackage[T1]{fontenc}
963 % Math support. amsmath is universal and works on every engine, so the class
964 % brings it in unconditionally (it must precede hyperref, which is loaded at
965 % the end of the class). amssymb is the legacy AMS symbol package, built
966 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
967 % XeLaTeX) it clashes with unicode-math, which already provides all of its
968 % symbols through Unicode math codes. So amssymb is loaded ONLY under
969 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
970 % unicode-math (the class does not auto-load unicode-math because choosing
971 % an OpenType math font is a typographic decision that belongs to the author).
972 \RequirePackage{amsmath}
973 \ifPDFTeX
974   \RequirePackage{amssymb}
975 \else
976   % Unicode engine: defer the check to end of document. Only warn if the
977   % user did NOT add either unicode-math (the modern OpenType-math route)
978   % or amssymb (the legacy route, which they would have loaded manually).
979   % This avoids nagging authors who already followed the documented
980   % \usepackage{unicode-math} workflow.
981   \AtEndDocument{%
982     \@ifpackageloaded{unicode-math}{\{%
983       \@ifpackageloaded{amssymb}{\}%
984       \ClassWarningNoLine{coppe}{%
985         You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
986         neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
987         (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
988         Add to your preamble:\MessageBreak
989         \space\space\string\usepackage{unicode-math}\MessageBreak
990         \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
991         Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
992         with the OpenType math font setup}}}%
993 \fi
994 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
995 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.

```

```

996 % headheight=15pt is given here (not patched later) so geometry's own page
997 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
998 % height to host the running head, and the standard 12pt default triggers
999 % its "headheight is too small" warning. Letting geometry know up front
1000 % keeps \topmargin a derived value -- we never touch it by hand.
1001 \RequirePackage[a4paper,twoside,bindingoffset=1cm,headheight=15pt,%
1002 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
1003 \RequirePackage{titlesec}
1004 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1005 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitolo N").
1006 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1007 % keep "N Title". \appendix and \annex raise the flag (see below).
1008 \newif\if@coppeappendix \@coppeappendixfalse
1009 \titleformat{\chapter}[hang]
1010   {\normalfont\Large\bfseries}
1011   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1012   {1ex}}{\MakeUppercase}
1013 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1014 \let\coppe@orig@appendix\appendix
1015 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1016 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1017 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1018 % which macro is written (so it is correct at .toc-render time even though the
1019 % appendixname has changed back); the chapter letter is baked in at write time.
1020 \newif\if@coppeannex \@coppeannexfalse
1021 \newcommand\coppe@tocapp[1]{%
1022   \MakeUppercase{\coppestring{appendix}}%
1023   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1024 \newcommand\coppe@tocanx[1]{%
1025   \MakeUppercase{\coppestring{annex}}%
1026   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1027 \def\@chapter[#1]#2{%
1028   \ifnum \c@secnumdepth >\m@ne
1029     \if@mainmatter
1030       \refstepcounter{chapter}%
1031       \typeout{\@chapapp\space\thechapter.}%
1032       \if@coppeappendix
1033         \if@coppeannex
1034           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1035         \else
1036           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1037         \fi
1038       \else
1039         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1040       \fi
1041     \else
1042       \addcontentsline{toc}{chapter}{#1}%
1043     \fi
1044   \else
1045     \addcontentsline{toc}{chapter}{#1}%
1046   \fi
1047   \chaptermark{#1}%
1048   \addtocontents{lof}{\protect\addvspace{10\p@}}%
1049   \addtocontents{lot}{\protect\addvspace{10\p@}}%

```

```

1050 \if@twocolumn
1051 \topnewpage[\@makechapterhead{#2}]%
1052 \else
1053 \@makechapterhead{#2}%
1054 \afterheading
1055 \fi}
1056 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1057 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1058 \titleformat{name=\chapter,numberless}[hang]
1059 {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1060 \titlespacing{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1061 % section: ALL CAPS, normal weight.
1062 \titleformat{section}
1063 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1064 % subsection: bold (author writes Title Case).
1065 \titleformat{subsection}
1066 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1067 % subsubsection: bold italic.
1068 \titleformat{subsubsection}
1069 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1070 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1071 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1072 \RequirePackage{caption}
1073 \captionsetup{font=footnotesize,labelsep=endash}
1074 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1075 % caption so it picks up the ABNT captionsetup above.
1076 \RequirePackage{subcaption}
1077 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1078 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1079 % at the bottom is mandatory). The float package's plaintop style forces the
1080 % caption above the body for figure, table and the new quadro float. -----
1081 \RequirePackage{float}
1082 \floatstyle{plaintop}
1083 \restylefloat{figure}
1084 \restylefloat{table}
1085 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1086 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1087 % unlike a tabela (rules on top and bottom only).
1088 \providecommand{\quadroname}{\coppestring{frame}}
1089 \providecommand{\listquadroname}{\coppestring{listframe}}
1090 \newfloat{quadro}{htbp}{loq}[chapter]
1091 \floatname{quadro}{\quadroname}
1092 \providecommand{\quadroautorefname}{\quadroname}
1093 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1094 % (float's own \l@quadro lays the number/title out incorrectly).
1095 \let\l@quadro\l@figure
1096 \newcommand\listofquadros{%
1097 \chapter*{\listquadroname}%
1098 \addcontentsline{toc}{chapter}{\listquadroname}%
1099 \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1100 \starttoc{loq}}
1101 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1102 % float as a smaller, single-spaced, centred line; it does NOT advance any
1103 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);

```

```

1104 % the label is "Fonte" (pt) / "Source" (en).
1105 \newcommand{\cpsourcenam}{\coppestring{source}}
1106 \newcommand{\cpsource}[1]{%
1107   \par\addvspace{\smallskipamount}%
1108   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1109 \let\cpfonte\cpsource
1110 \@ifundefined{source}{\let\source\cpsource}{}%
1111 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1112 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1113 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1114 \newcommand{\newcoppefloat}[3]{%
1115   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1116   \floatname{#1}{#2}%
1117   \expandafter\newcommand\csname listof#1s\endcsname{%
1118     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1119     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1120     \@starttoc{lo#1}}}%
1121 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1122 %   (filete) from the left margin; footnotesize is the book default. -----
1123 \let\coppe@orig@footnotetext\@footnotetext
1124 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1125 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1126 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1127 %   (manual2024 2.6 / proposta P24). -----
1128 \RequirePackage{enumitem}
1129 \newlist{alineas}{enumerate}{2}
1130 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1131 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1132 % --- Page numbering: arabic in the top OUTER corner on textual pages
1133 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1134 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1135 %   class option). -----
1136 \RequirePackage{fancyhdr}
1137 % \headheight is set to 15pt by geometry up above (see the geometry call),
1138 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1139 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}
1140 \renewcommand{\headrulewidth}{0pt}}
1141 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1142 % number may leak onto them (e.g. the verso right after the cover).
1143 \renewcommand{\cleardoublepage}{%
1144   \clearpage
1145   \if@twoside
1146     \ifodd\c@page\else
1147       \hbox{}\thispagestyle{empty}\newpage
1148     \if@twocolumn\hbox{}\newpage\fi
1149   \fi
1150   \fi}
1151 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1152 p\kern-.025em e}}\kern-.08em
1153 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1154 \newboolean{maledoc}
1155 \setboolean{maledoc}{false}
1156 %
1157 % Class options.

```



```

1158 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1159 % fixed language slots:
1160 % * main      -- the language of the body, captions, and the main abstract.
1161 %              Set by a class option: brazilian (default), english,
1162 %              spanish, french, italian, or any user-supplied language
1163 %              that ships a coppe-lang-<name>.def file.
1164 % * foreign   -- the language used by the foreignabstract environment and
1165 %              by \foreign@... registers. Defaults to english (or to
1166 %              brazilian when main=english) for back-compat.
1167 % * third     -- optional, loaded on demand with \usecoppelanguage{<name>}
1168 %              in the preamble (for citations or quotations in a third
1169 %              language). Has no class option of its own.
1170 % Note: the legacy \if@english boolean (defined and toggled here in
1171 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1172 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1173 % tests it. Code that needs to branch on the main language should test
1174 % \ifx\coppe@mainlang\coppe@englishstr instead.
1175 \def\coppe@mainlang{brazilian}
1176 \def\coppe@foreignlang{english}
1177 \DeclareOption{brazilian}{%
1178   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1179 \DeclareOption{english}{%
1180   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1181 \DeclareOption{spanish}{%
1182   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1183 \DeclareOption{french}{%
1184   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1185 \DeclareOption{italian}{%
1186   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1187 \DeclareOption{msc}{%
1188   \newcommand{\@degree}{M.Sc.}
1189   \newcommand{\@degreename}{Mestrado}
1190   \newcommand{\local@degname}{Mestre}
1191   \newcommand{\foreign@degname}{Master}
1192   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1193   \newcommand\foreign@doctype{Dissertation}
1194 }
1195 \DeclareOption{dscexam}{%
1196   \newcommand{\@degree}{D.Sc.}
1197   \newcommand{\@degreename}{Doutorado}
1198   \newcommand{\local@degname}{Doutor}
1199   \newcommand{\foreign@degname}{Doctor}
1200   \setboolean{maledoc}{true}
1201   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1202   \newcommand\foreign@doctype{Qualifying Exam}
1203 }
1204 \DeclareOption{mscexam}{%
1205   \newcommand{\@degree}{M.Sc.}
1206   \newcommand{\@degreename}{Mestrado}
1207   \newcommand{\local@degname}{Mestre}
1208   \newcommand{\foreign@degname}{Master}
1209   \setboolean{maledoc}{true}
1210   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1211   \newcommand\foreign@doctype{Qualifying Exam}

```

```

1212 }
1213 \DeclareOption{dsc}{%
1214   \newcommand{\@degree}{D.Sc.}
1215   \newcommand{\@degreename}{Doutorado}
1216   \newcommand{\local@degname}{Doutor}
1217   \newcommand{\foreign@degname}{Doctor}
1218   \newcommand{\local@doctype}{Tese}
1219   \newcommand{\foreign@doctype}{Thesis}
1220 }
1221 \newif\if@coppenumeric\@coppenumericfalse
1222 \DeclareOption{numbers}{\@coppenumerictrue}
1223 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1224 % pre-textual pages are counted but not numbered (manual2024 2.7).
1225 \newif\if@copperoman\@copperomanfalse
1226 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1227 \newif\if@coppedoublespace\@coppedoublespacefalse
1228 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1229 \ProcessOptions\relax
1230 % Built-in language names (used by the loader and the dispatcher below).
1231 \def\coppe@@brazilianstr{brazilian}
1232 \def\coppe@@englishstr{english}
1233 % Load Babel with the foreign language first and the main language last
1234 % (Babel treats the LAST option in the list as the main language). With
1235 % pt-main this produces [english,brazilian]{babel} and with en-main
1236 % [brazilian,english] -- identical to the historical \if@english calls.
1237 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1238 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1239 % --- Multilingual string dispatcher -----
1240 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1241 % language <lang>. The three retrievers below look strings up either at the
1242 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1243 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1244 \providecommand{\copperdefstring}[3]{%
1245   \expandafter\gdef\csname coppe@str@#1#2\endcsname{#3}}
1246 \providecommand{\coppestring}[1]{\@nameuse{coppe@str\@languagename @#1}}
1247 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str\coppe@mainlang @#1}}
1248 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str\coppe@foreignlang @#1}}
1249 % Load an extra language pack at any point in the preamble (third-language
1250 % slot for citations or quotations beyond the main+foreign pair).
1251 \providecommand{\usecoppelanguage}[1]{%
1252   \def\coppe@@cur{#1}%
1253   \ifx\coppe@@cur\coppe@@brazilianstr\else
1254   \ifx\coppe@@cur\coppe@@englishstr\else
1255     \InputIfFileExists{coppe-lang-#1.def}{%
1256       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1257     }{\fi}
1258 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1259 % consulted by the cover/folha renderers once they are migrated in step 2.
1260 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle

```

```

1261 % still drive every visible site, so behavior for pt/en is unchanged.
1262 \providecommand{\titlein}[2]{%
1263   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1264 % --- Built-in string tables: brazilian + english. The values mirror the
1265 % literals currently scattered through the class; step-1 only populates the
1266 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1267 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1268 \copperdefstring{english}{appendix}{Appendix}
1269 \copperdefstring{brazilian}{annex}{Anexo}
1270 \copperdefstring{english}{annex}{Annex}
1271 \copperdefstring{brazilian}{frame}{Quadro}
1272 \copperdefstring{english}{frame}{Frame}
1273 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1274 \copperdefstring{english}{listframe}{List of Frames}
1275 \copperdefstring{brazilian}{source}{Fonte}
1276 \copperdefstring{english}{source}{Source}
1277 \copperdefstring{brazilian}{program}{Programa}
1278 \copperdefstring{english}{program}{Program}
1279 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1280 \copperdefstring{english}{listprogram}{List of Programs}
1281 \copperdefstring{brazilian}{references}{Refer\~encias}
1282 \copperdefstring{english}{references}{References}
1283 \copperdefstring{brazilian}{in:}{em}
1284 \copperdefstring{english}{in:}{in}
1285 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1286 \copperdefstring{english}{art}{Art}
1287 \copperdefstring{brazilian}{platform}{Plataforma}
1288 \copperdefstring{english}{platform}{Platform}
1289 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1290 \copperdefstring{english}{directedby}{Directed by}
1291 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1292 \copperdefstring{english}{producedby}{Produced by}
1293 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1294 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1295 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~i}mbolos}
1296 \copperdefstring{english}{listsymbol}{List of Symbols}
1297 \copperdefstring{brazilian}{glossary}{Gloss{\~a}rio}
1298 \copperdefstring{english}{glossary}{Glossary}
1299 \copperdefstring{brazilian}{advisor}{Orientador}
1300 \copperdefstring{english}{advisor}{Advisor}
1301 \copperdefstring{brazilian}{advisors}{Orientadores}
1302 \copperdefstring{english}{advisors}{Advisors}
1303 \copperdefstring{brazilian}{dept}{Programa}
1304 \copperdefstring{english}{dept}{Department}
1305 \copperdefstring{brazilian}{approvedmale}{Examinado por}
1306 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1307 \copperdefstring{english}{approvedmale}{Examined by}
1308 \copperdefstring{english}{approvedfemale}{Examined by}
1309 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1310 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1311 \copperdefstring{english}{universityname}{Universidade Federal do Rio de Janeiro}
1312 \copperdefstring{brazilian}{cityname}{Rio de Janeiro}
1313 \copperdefstring{english}{cityname}{Rio de Janeiro}
1314 \copperdefstring{brazilian}{statename}{RJ}

```

```

1315 \copperdefstring{english} {statename} {RJ}
1316 \copperdefstring{brazilian}{countryname} {Brasil}
1317 \copperdefstring{english} {countryname} {Brasil}
1318 % Month names.
1319 \copperdefstring{brazilian}{monthname1} {Janeiro}
1320 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1321 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1322 \copperdefstring{brazilian}{monthname4} {Abril}
1323 \copperdefstring{brazilian}{monthname5} {Maio}
1324 \copperdefstring{brazilian}{monthname6} {Junho}
1325 \copperdefstring{brazilian}{monthname7} {Julho}
1326 \copperdefstring{brazilian}{monthname8} {Agosto}
1327 \copperdefstring{brazilian}{monthname9} {Setembro}
1328 \copperdefstring{brazilian}{monthname10}{Outubro}
1329 \copperdefstring{brazilian}{monthname11}{Novembro}
1330 \copperdefstring{brazilian}{monthname12}{Dezembro}
1331 \copperdefstring{english}{monthname1} {January}
1332 \copperdefstring{english}{monthname2} {February}
1333 \copperdefstring{english}{monthname3} {March}
1334 \copperdefstring{english}{monthname4} {April}
1335 \copperdefstring{english}{monthname5} {May}
1336 \copperdefstring{english}{monthname6} {June}
1337 \copperdefstring{english}{monthname7} {July}
1338 \copperdefstring{english}{monthname8} {August}
1339 \copperdefstring{english}{monthname9} {September}
1340 \copperdefstring{english}{monthname10}{October}
1341 \copperdefstring{english}{monthname11}{November}
1342 \copperdefstring{english}{monthname12}{December}
1343 % algorithm2e package option for this language (empty -> english default).
1344 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1345 \copperdefstring{english} {algorithm2eopt}{}
1346 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1347 \copperdefstring{brazilian}{babelname}{brazilian}
1348 \copperdefstring{english} {babelname}{english}
1349 % Auto-load language packs for any non-built-in language listed in babelopts.
1350 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1351 % (defined just above) and need no extra file; others (spanish/french/italian
1352 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1353 % must already be defined when these run, so the loader sits AFTER the tables.
1354 \@for\coppe@@lang:=\coppe@babelopts\do{%
1355   \edef\coppe@@cur{\coppe@@lang}%
1356   \ifx\coppe@@cur\coppe@@brazilianstr\else
1357     \ifx\coppe@@cur\coppe@@englishstr\else
1358       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1359       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1360       {\ClassError{coppe}
1361        {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1362        {Install it next to coppe.cls, or pick a built-in language.}}%
1363     \fi\fi}
1364 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1365 % Portuguese alias for short, inline quotations (proposta P3).
1366 \RequirePackage{csquotes}
1367 \newcommand{\citacao}[1]{\enquote{#1}}
1368 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings

```

```

1369 %      package gives the "Programa" listing (caption on top, ABNT R46) with
1370 %      \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1371 %      the language styles below are based on the user's listagens.tex. -----
1372 \RequirePackage{xcolor}
1373 \RequirePackage{listings}
1374 \RequirePackage{fancyvrb}
1375 \DeclareFixedFont{\ttb}{T1}{ttxtt}{bx}{n}{10}
1376 \DeclareFixedFont{\ttm}{T1}{ttxtt}{m}{n}{10}
1377 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1378   literate=%
1379   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1380   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1381   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1382   {\~}{\~{C}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1383   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{U}}1%
1384   {\~}{\~{U}}1%
1385 }
1386 % Babel-aware listing names, set after listings' own babel setup.
1387 \AtBeginDocument{%
1388   \renewcommand\lstlistingname{\coppestring{program}}%
1389   \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1390 % \listofprogramas in the coppe list style (heading + ToC + running head).
1391 \newcommand\listofprogramas{%
1392   \chapter*{\lstlistlistingname}%
1393   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1394   \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1395   \@starttoc{lol}}
1396 % Per-language styles + environments (python, java, xml, html, prolog).
1397 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1398   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1399   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1400   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1401   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1402   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1403   escapechar=',postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1404 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}]%
1405   \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1406   \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!]
1407 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]
1408 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1409   numberstyle=\small\ttfamily,numbersep=0.5em,%
1410   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1411   keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1412   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1413   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1414   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1415   postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1416 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]
1417 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1418   numberstyle=\small\ttfamily,numbersep=0.5em,%
1419   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1420   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1421   extendedchars=true,inputencoding=utf8,breaklines=true,%
1422   postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}

```

```

1423 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1424 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1425   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1426   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1427   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1428   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1429   breaklines=true,escapechar=',%
1430   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}
1431 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1432   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}%
1433 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1434   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1435   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1436   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1437   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1438   breaklines=true,escapechar=',%
1439   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}
1440 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1441   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}%
1442 \newcommand\javainline[1]{\javastyle\lstinline!#1!}%
1443 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}]%
1444 % fancyvrb's Verbatim environment is available for program output; the
1445 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1446 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1447 % example,
1448 %   \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1449 %     ...
1450 %   \end{Verbatim}
1451 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1452 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
1453 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1454 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1455 % language (English) -- needed because algorithm2e errors on an empty option.
1456 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1457 \ifx\coppe@@algotlang\empty
1458   \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1459 \else
1460   \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1461 \fi
1462 \SetKwFor{Para}{para}{faça}{fim do para}
1463 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1464 % setspace + hyperref + biblatex are loaded at the very end of the class
1465 % (see the block just before %</class>), so their hooks see every macro the
1466 % class defines, and the load order setspace -> hyperref -> biblatex is
1467 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1468 \newcommand\department[1]{%
1469   \ifthenelse{\equal{#1}{PEB}}{
1470     {\global\def\local@deptname{Engenharia Biom(' e)dica}
1471     \global\def\foreign@deptname{Biomedical Engineering}}{
1472     \ifthenelse{\equal{#1}{PEC}}{

```

```

1473     {\global\def\local@deptname{Engenharia Civil}
1474     \global\def\foreign@deptname{Civil Engineering}}{}
1475 \ifthenelse{\equal{#1}{PEE}}
1476     {\global\def\local@deptname{Engenharia El{\` e}trica}
1477     \global\def\foreign@deptname{Electrical Engineering}}{}
1478 \ifthenelse{\equal{#1}{PEM}}
1479     {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1480     \global\def\foreign@deptname{Mechanical Engineering}}{}
1481 \ifthenelse{\equal{#1}{PEMM}}
1482     {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1483 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1484 \ifthenelse{\equal{#1}{PEN}}
1485     {\global\def\local@deptname{Engenharia Nuclear}
1486     \global\def\foreign@deptname{Nuclear Engineering}}{}
1487 \ifthenelse{\equal{#1}{PEN0}}
1488     {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1489     \global\def\foreign@deptname{Ocean Engineering}}{}
1490 \ifthenelse{\equal{#1}{PPE}}
1491     {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1492     \global\def\foreign@deptname{Energy Planning}}{}
1493 \ifthenelse{\equal{#1}{PEP}}
1494     {\global\def\local@deptname{Engenharia de Produ{\c c}{\` a}o}
1495     \global\def\foreign@deptname{Production Engineering}}{}
1496 \ifthenelse{\equal{#1}{PEQ}}
1497     {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1498     \global\def\foreign@deptname{Chemical Engineering}}{}
1499 \ifthenelse{\equal{#1}{PESC}}
1500     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\` a}o}
1501     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1502 \ifthenelse{\equal{#1}{PET}}
1503     {\global\def\local@deptname{Engenharia de Transportes}
1504     \global\def\foreign@deptname{Transportation Engineering}}{}
1505 \ifthenelse{\equal{#1}{PENT}}
1506     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1507     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1508 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1509 \renewcommand\title[1]{%
1510 \global\def\local@title{#1}%
1511 \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1512 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1513 \newcommand\foreigntitle[1]{%
1514 \global\def\foreign@title{#1}%
1515 \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1516 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is

\local@title; for en-main \foreign@title; for any other main language the user must call \titlein{<lang>}{...} to populate it.

```
1517 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
```

\advisor Defines globally the title, name and academic degree of the advisors.

```
1518 \newcount\@advisor\@advisor0
1519 \newcommand\advisor[4]{%
1520   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1521   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1522   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1523   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1524   \global\advance\@advisor by 1
1525   \ifnum\@advisor>1
1526     \renewcommand\local@advisorstring{Orientadores}
1527     \renewcommand\foreign@advisorstring{Advisors}
1528   \fi
1529 }
```

\examiner

```
1530 \newcount\@examiner\@examiner0
1531 \newcommand\examiner[3]{%
1532   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1533   \global\advance\@examiner by 1
1534 }
```

\author It was redefined to allow the identification of the author's first names and surname.

```
1535 \renewcommand\author[2]{%
1536   \global\def\@authname{#1}
1537   \global\def\@authsurn{#2}
1538 }
```

\date This code makes easy to switch from dates in different languages.

```
1539 \renewcommand\date[2]{%
1540   \month=#1
1541   \year=#2
1542 }
```

\local@monthname

```
1543 \newcommand\local@monthname{\ifcase\month\or
1544   Janeiro\or Fevereiro\or Mar{\c c}\or Abril\or Maio\or Junho\or
1545   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

\foreign@monthname

```
1546 \newcommand\foreign@monthname{\ifcase\month\or
1547   January\or February\or March\or April\or May\or June\or
1548   July\or August\or September\or October\or November\or December\fi}
```

\keyword

```
1549 \newcounter{keywords}
1550 \newcommand\keyword[1]{%
1551   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1552   \global\addtocounter{keywords}{1}
1553 }
```


`\freeconfig` This command allows easy changing of core class parameters.

```

1554 \newcommand\freeconfig[6]{%
1555 \providecommand\@degree{}
1556 \renewcommand\@degree{#1}
1557 \providecommand\@degreename{}
1558 \renewcommand\@degreename{#2}
1559 \providecommand\local@degname{}
1560 \renewcommand\local@degname{#3}
1561 \providecommand\foreign@degname{}
1562 \renewcommand\foreign@degname{#4}
1563 \providecommand\local@doctype{}
1564 \renewcommand\local@doctype{#5}
1565 \providecommand\foreign@doctype{}
1566 \renewcommand\foreign@doctype{#6}%
1567 }
1568 % \end{macrocode}
1569 % \end{macro}
1570 %
1571 % \begin{macro}{\frontmatter}
1572 % The number of pages for both frontmatter and mainmatter printed
1573 % in the cataloging details page is computed by means of simple
1574 % \LaTeX\ labels.
1575 % \begin{macrocode}
1576 \renewcommand\frontmatter{%
1577 \cleardoublepage
1578 \@mainmatterfalse
1579 \pagenumbering{roman}
1580 \thispagestyle{empty}
1581 % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1582 % start keeps the page-counter parity aligned with the physical (recto) side,
1583 % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1584 \setcounter{page}{1}
1585 \makefrontpage
1586 \clearpage
1587 \if@copperoman
1588 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1589 \renewcommand{\headrulewidth}{0pt}}%
1590 \pagestyle{coppe}%
1591 \else
1592 \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1593 \pagestyle{empty}%
1594 \fi
1595 }
```

`\mainmatter`

```

1596 \renewcommand\mainmatter{%
1597 \coppe@mainBegin
1598 \cleardoublepage
1599 \@mainmattertrue
1600 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1601 \renewcommand{\headrulewidth}{0pt}}%
1602 \pagestyle{coppe}%
1603 \pagenumbering{arabic}}
```

`\backmatter`

```
1604 \renewcommand\backmatter{%
1605   \if@openright
1606     \cleardoublepage
1607   \else
1608     \clearpage
1609   \fi}
1610 %
```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```
1611 \newcommand\makecover{%
1612   \begin{titlepage}%
1613   \centering
1614   \makebox[\linewidth]{%
1615     \includegraphics[height=2cm]{ufrj-logo}\hfill
1616     \includegraphics[height=2cm]{coppe-logo}}\par
1617   \vspace{1.5cm}
1618   {\singlespacing
1619     \MakeUppercase{\local@universityname}\par
1620     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1621       P{\~{o}}s-Gradua{\c c}{\~{a}}o e Pesquisa de Engenharia (COPPE)}}\par
1622     \MakeUppercase{Programa de \local@deptname}\par}
1623   \vspace{\stretch{2}}
1624   \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1625   \vspace{\stretch{2}}
1626   \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1627   \vspace{\stretch{3}}
1628   \MakeUppercase{\local@cityname}\par
1629   \number\year
1630   \end{titlepage}%
1631 }
```

`\maketitle`

```
1632 \renewcommand\maketitle{%
1633   \pagenumbering{alph}
1634   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1635   \makecover
1636   \begin{titlepage}
1637   \begin{flushleft}
1638     \vspace*{1.5mm}
1639     \setlength\baselineskip{0pt}
1640     \setlength\parskip{1mm}
1641     \makebox[\linewidth]{%
1642       \includegraphics[height=2cm]{ufrj-logo}\hfill
1643       \includegraphics[height=2cm]{coppe-logo}}
1644     \end{flushleft}
1645     \vspace{1.05cm}
1646     \begin{center}
```

```

1647 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1648 \vspace*{3cm}
1649 \nohyphens{\@authname\ \@authsurn}\par
1650 \end{center}
1651 \vspace*{2.1cm}
1652 \begin{flushright}
1653 \begin{minipage}{8.45cm}
1654 \frontcover@maintext
1655 \end{minipage}\par
1656 \vspace*{7.5mm}
1657 \nohyphens{%
1658 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1659 \local@advisorstring: &
1660 \count1=0
1661 \toks@={}
1662 \@whilenum \count1<\@advisor \do{%
1663 \ifcase\count1 % same as \ifnum0=\count1
1664 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1665 \expandafter\endcsname\expandafter\space%
1666 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1667 \else
1668 \toks@=\expandafter\expandafter\expandafter{%
1669 \expandafter\the\expandafter\toks@%
1670 \expandafter&\expandafter\space%
1671 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1672 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1673 }%
1674 \fi
1675 \advance\count1 by 1}
1676 \the\toks@
1677 \end{tabularx}}\par
1678 \end{flushright}
1679 \vspace*{\fill}
1680 \begin{center}
1681 \local@cityname\par
1682 \local@monthname\ de \number\year
1683 \end{center}
1684 \end{titlepage}
1685 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1686 % the title page is a recto, so this lands on the immediately following verso,
1687 % ahead of the approval sheet produced by \frontmatter.
1688 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1689 \global\let\maketitle\relax%
1690 \global\let\and\relax}

1691 \newcommand\makefrontpage{%
1692 \begin{center}
1693 \sloppy\nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1694 \vspace*{7mm}
1695 {\@authname\ \@authsurn}\par
1696 \end{center}\par
1697 \vspace*{4mm}
1698 \frontpage@maintext
1699 \vspace*{16mm}

```

```

1700 \nohyphens{%
1701 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}}
1702 \local@advisorstring: &
1703 \count1=0
1704 \toks@={}
1705 \@whilenum \count1<\@advisor \do{%
1706 \ifcase\count1 % same as \ifnum0=\count1
1707 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1708 \expandafter\endcsname\expandafter\space%
1709 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1710 \else
1711 \toks@=\expandafter\expandafter\expandafter{%
1712 \expandafter\the\expandafter\toks@%
1713 \expandafter&\expandafter\space%
1714 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1715 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1716 }%
1717 \fi
1718 \advance\count1 by 1}
1719 \the\toks@
1720 \end{tabularx}\par
1721 \vspace*{20mm}
1722 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}}
1723 Aprovada por: &
1724 \count1=0
1725 \toks@={}
1726 \@whilenum \count1<\@examiner \do{%
1727 \ifcase\count1 % same as \ifnum0=\count1
1728 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1729 \expandafter\endcsname\expandafter\\}
1730 \else
1731 \toks@=\expandafter\expandafter\expandafter{%
1732 \expandafter\the\expandafter\toks@%
1733 \expandafter&\expandafter\space%
1734 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1735 \expandafter\space\\}
1736 }%
1737 \fi
1738 \advance\count1 by 1}
1739 \the\toks@
1740 \end{tabularx}}\par
1741 \vspace*{\fill}
1742 \frontpage@bottomtext}

1743 \newcommand\coppe@hypersetup{%
1744 \begin{group}
1745 % changes to \toks@ and \count@ are kept local;
1746 % it's not necessary for them, but it is usually the case
1747 % for \count1, because the first ten counters are written
1748 % to the DVI file, thus you got lucky because of PDF output
1749 \toks@={} % in this special case not necessary
1750 \count@=0 %
1751 \@whilenum\count@<\value{keywords}\do{%
1752 % * a keyword separator is not necessary,

```

```

1753 % if there is just one keyword
1754 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1755 % at least once, to get rid of the loop depended \count@
1756 \ifcase\count@ % same as \ifnum0=\count@
1757 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1758 \else
1759 \toks@=\expandafter\expandafter\expandafter{%
1760 \expandafter\the\expandafter\toks@
1761 \expandafter;\expandafter\space
1762 \csname CoppeKeyword:\the\count@\endcsname
1763 }%
1764 \fi
1765 \advance\count@ by 1 %
1766 }%
1767 \edef\x{\endgroup
1768 \noexpand\hypersetup{%
1769 pdfkeywords={\the\toks@}%
1770 }%
1771 }%
1772 \x
1773 \hypersetup{%
1774 pdfauthor={\@authname\ \@authsurn},
1775 pdftitle={\local@title},
1776 pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
1777 pdfcreator={LaTeX with CoppeTeX toolkit},
1778 breaklinks={true},
1779 raiselinks={true},
1780 pageanchor={true},
1781 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain $\text{\TeX}$ command `\@isundefined` (cf. the $\text{\TeX}$ FAQ).

```

1782 \newcommand\makecatalog{%
1783 \vspace*{\fill}
1784 \begin{center}
1785 \setlength{\fboxsep}{5mm}
1786 \framebox[120mm][c]{\makebox[5mm][c]{}%
1787 \begin{minipage}[c]{105mm}
1788 \setlength{\parindent}{5mm}
1789 \noindent\sloppy\nohyphens\@authsurn,
1790 \nohyphens\@authname\par
1791 \nohyphens{%
1792 \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
1793 UFRJ/COPPE, \number\year.}\par
1794 \pageref{front:pageno},
1795 \pageref{LastPage}
1796 p.\@ifundefined{r@cat:lofflag}{}{\pageref{cat:lofflag}} $29,7$cm.\par
1797 % There is an issue here. When the last entry must be split between lines,
1798 % the spacing between it and the next paragraph becomes smaller.
1799 % Should we manually introduce a fixed space? But how could we know that
1800 % a name was split? Is this happening yet?
1801 \nohyphens}%

```

```

1802 \begin{tabularx}{100mm}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1803 \local@advisorstring: &
1804 \count1=0
1805 \toks@={ }
1806 \@whilenum \count1<\@advisor \do{%
1807 \ifcase\count1 % same as \ifnum0=\count1
1808 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1809 \expandafter\endcsname\expandafter\space%
1810 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1811 \else
1812 \toks@=\expandafter\expandafter\expandafter{%
1813 \expandafter\the\expandafter\toks@
1814 \expandafter&\expandafter\space
1815 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1816 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1817 }%
1818 \fi
1819 \advance\count1 by 1}
1820 \the\toks@
1821 \end{tabularx}}\par
1822 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1823 UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1824 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1825 \count1=0
1826 \count2=1
1827 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1828 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1829 \advance\count1 by 1
1830 \advance\count2 by 1}
1831 I. \csname CoppeAdvisorSurname:0\endcsname,%
1832 \ \csname CoppeAdvisorName:0\endcsname%
1833 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1834 II. \local@universityname, COPPE, Programa de \local@deptname.
1835 III. T{\' i}tulo.}
1836 \end{minipage}}
1837 \end{center}
1838 \vspace*{\fill}}

```

\dedication

```

1839 \newcommand\dedication[1]{
1840 \gdef\@dedic{#1}
1841 \cleardoublepage
1842 \vspace*{\fill}
1843 \begin{flushright}
1844 \begin{minipage}{60mm}
1845 \raggedleft \it \normalsize \@dedic
1846 \end{minipage}
1847 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1848 \newenvironment{abstract}{%
1849 \cleardoublepage
1850 \thispagestyle{plain}
1851 \abstract@toptext\par

```

```

1852 \vspace*{8.6mm}
1853 \begin{center}
1854 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1855 \vspace*{13.2mm}
1856 \@authname\ \@authsur \par
1857 \vspace*{7mm}
1858 \local@monthname/\number\year
1859 \end{center}\par
1860 \vspace*{1.5cm}
1861 \noindent%
1862 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1863 \local@advisorstring: &
1864 \count1=0
1865 \toks@={}
1866 \@whilenum \count1<\@advisor \do{%
1867 \ifcase\count1 % same as \ifnum0=\count1
1868 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1869 \expandafter\endcsname\expandafter\space%
1870 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1871 \else
1872 \toks@=\expandafter\expandafter\expandafter{%
1873 \expandafter\the\expandafter\toks@
1874 \expandafter&\expandafter\space
1875 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1876 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1877 }%
1878 \fi
1879 \advance\count1 by 1}
1880 \the\toks@
1881 \end{tabularx}\par
1882 \vspace*{2mm}
1883 \noindent\local@deptstring: \local@deptname\par
1884 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (env.)

```

1885 \newenvironment{foreignabstract}{%
1886 \cleardoublepage
1887 \thispagestyle{plain}
1888 \begin{otherlanguage}{\coppe@foreignlang}
1889 \foreignabstract@toptext\par
1890 \vspace*{8.6mm}
1891 \begin{center}
1892 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1893 \vspace*{13.2mm}
1894 \@authname\ \@authsur \par
1895 \vspace*{7mm}
1896 \foreign@monthname/\number\year
1897 \end{center}\par
1898 \vspace*{1.5cm}
1899 \noindent%
1900 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1901 \foreign@advisorstring: &
1902 \count1=0
1903 \toks@={}

```

```

1904 \@whilenum \count1<\@advisor \do{%
1905 \ifcase\count1 % same as \ifnum0=\count1
1906 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1907 \expandafter\endcsname\expandafter\space%
1908 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1909 \else
1910 \toks@=\expandafter\expandafter\expandafter{%
1911 \expandafter\the\expandafter\toks@
1912 \expandafter&\expandafter\space
1913 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1914 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1915 }%
1916 \fi
1917 \advance\count1 by 1}
1918 \the\toks@
1919 \end{tabularx}\par
1920 \vspace*{2mm}
1921 \noindent\foreign@deptstring: \foreign@deptname\par
1922 \vspace*{7mm}}{%
1923 \end{otherlanguage}
1924 \vspace*{\fill}
1925 \global\let\@author\@empty
1926 \global\let\@date\@empty
1927 \global\let\foreign@title\@empty
1928 \global\let\foreign@title\relax
1929 \global\let\local@title\@empty
1930 \global\let\local@title\relax
1931 \global\let\author\relax
1932 \global\let\author\relax
1933 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

1934 \newenvironment{brazilianabstract}{%
1935 \cleardoublepage
1936 \thispagestyle{plain}
1937 \begin{otherlanguage}{brazilian}
1938 \abstract@toptext\par
1939 \vspace*{8.6mm}
1940 \begin{center}
1941 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1942 \vspace*{13.2mm}
1943 \@authname\ \@authsurn \par
1944 \vspace*{7mm}
1945 \local@monthname/\number\year
1946 \end{center}\par
1947 \vspace*{1.5cm}
1948 \noindent%

```



```

1949 \begin{tabularx}{\textwidth}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1950 \local@advisorstring: &
1951 \count1=0
1952 \toks@={}
1953 \@whilenum \count1<\@advisor \do{%
1954 \ifcase\count1
1955 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1956 \expandafter\endcsname\expandafter\space%
1957 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1958 \else
1959 \toks@=\expandafter\expandafter\expandafter{%
1960 \expandafter\the\expandafter\toks@
1961 \expandafter&\expandafter\space
1962 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1963 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1964 }%
1965 \fi
1966 \advance\count1 by 1}
1967 \the\toks@
1968 \end{tabularx}\par
1969 \vspace*{2mm}
1970 \noindent\local@deptstring: \local@deptname\par
1971 \vspace*{7mm}}{%
1972 \end{otherlanguage}
1973 \vspace*{\fill}}

```

\listoffigures

```

1974 \renewcommand\listoffigures{%
1975 \coppe@hasLof
1976 \if@twocolumn
1977 \@restonecoltrue\onecolumn
1978 \else
1979 \@restonecolfalse
1980 \fi
1981 \chapter*{\listfigurename}%
1982 \addcontentsline{toc}{chapter}{\listfigurename}%
1983 \@mkboth{\MakeUppercase\listfigurename}%
1984 {\MakeUppercase\listfigurename}%
1985 \@starttoc{lof}%
1986 \if@restonecol\twocolumn\fi
1987 }

```

\listoftables

```

1988 \renewcommand\listoftables{%
1989 \if@twocolumn
1990 \@restonecoltrue\onecolumn
1991 \else
1992 \@restonecolfalse
1993 \fi
1994 \chapter*{\listtablename}%
1995 \addcontentsline{toc}{chapter}{\listtablename}%
1996 \@mkboth{%
1997 \MakeUppercase\listtablename}%
1998 {\MakeUppercase\listtablename}%

```

```

1999 \starttoc{lot}%
2000 \if@restonecol\twocolumn\fi
2001 }

\printlosymbols

2002 \newcommand\printlosymbols{%
2003 \renewcommand\glossaryname{\listsymbolname}%
2004 \@input@{\jobname.los}}

\makelosymbols

2005 \def\makelosymbols{%
2006 \newwrite\@losfile
2007 \immediate\openout\@losfile=\jobname.syx
2008 \newcommand\syml[3][\@bsphack\begingroup
2009 \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2010 \@sanitize%
2011 \@wrtos{\@tempsymb1}{##2}{##3}\typeout%
2012 {Writing index of symbols file \jobname.syx}%
2013 \let\makelosymbols\@empty%
2014 }%
2015 \@onlypreamble\makelosymbols

2016 \AtBeginDocument{%
2017 \ifpackageloaded{hyperref}{%
2018 \newcommand\@wrtos[3]{%
2019 \protected@write\@losfile{%
2020 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2021 \endgroup%
2022 \@esphack}}{%
2023 \newcommand\@wrtos[3]{%
2024 \protected@write\@losfile{%
2025 {\string\indexentry{#1[#2] #3}{\thepage}}%
2026 \endgroup%
2027 \@esphack}}}%

\printloabbreviations

2028 \newcommand\printloabbreviations{%
2029 \renewcommand\glossaryname{\listabbreviationname}%
2030 \@input@{\jobname.lab}}

\makeloabbreviations

2031 \def\makeloabbreviations{%
2032 \newwrite\@labfile
2033 \immediate\openout\@labfile=\jobname.abx
2034 \newcommand\abbrev[3][\@bsphack\begingroup
2035 \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2036 \@sanitize%
2037 \@wrtlab{\@tempsymb1}{##2}{##3}\typeout%
2038 {Writing index of abbreviations file \jobname.abx}%
2039 \let\makeloabbreviations\@empty
2040 }
2041 \@onlypreamble\makeloabbreviations

```

```

2042 \AtBeginDocument{%
2043 \ifpackageloaded{hyperref}{%
2044 \newcommand\wrlab[3]{%
2045 \protected@write\@labfile{%
2046 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2047 \endgroup%
2048 \esphack}}{%
2049 \newcommand\wrlab[3]{%
2050 \protected@write\@labfile{%
2051 {\string\indexentry{#1[#2] #3}{\thepage}}%
2052 \endgroup%
2053 \esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2054 \newcommand\newsigla[3]{%
2055 \namedef{coppe@sigla@s#1}{#2}%
2056 \namedef{coppe@sigla@l#1}{#3}%
2057 \newcommand\coppe@sigla@reg[1]{%
2058 \ifundefined{abbrev}{}%
2059 {\abbrev{\@nameuse{coppe@sigla@s#1}}{\@nameuse{coppe@sigla@l#1}}}%
2060 \newcommand\coppe@sigla@full[1]{%
2061 \@nameuse{coppe@sigla@l#1}\space(\@nameuse{coppe@sigla@s#1})}
2062 \newcommand\coppe@sigla@undef[1]{%
2063 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2064 \textbf{???#1???}}
2065 \newcommand\coppe@sigla@first[1]{%
2066 \global\@namedef{coppe@sigla@done#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2067 \newcommand\coppe@sigla@norm[1]{%
2068 \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2069 {\@ifundefined{coppe@sigla@done#1}{\coppe@sigla@first{#1}}%
2070 {\@nameuse{coppe@sigla@s#1}}}%
2071 \newcommand\coppe@sigla@star[1]{%
2072 \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2073 {\global\@namedef{coppe@sigla@done#1}{\coppe@sigla@full{#1}}}%
2074 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2075 %%% \AtBeginDocument{%
2076 %%% \ifpackageloaded{hyperref}{%
2077 %%% \def\wrlab#1#2{%
2078 %%% \protected@write\@labfile{%
2079 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2080 %%% \endgroup
2081 %%% \esphack}}{%
2082 %%% \def\wrlab#1#2{%
2083 %%% \protected@write\@labfile{%
2084 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2085 %%% \endgroup
2086 %%% \esphack}}

2087 % Some macros used to generate cataloging information.
2088 \AtBeginDocument{%

```

```

2089 \ltx@ifpackageloaded{hyperref}{
2090   \def\coppe@bibEnd{%
2091     \immediate\write\@auxout{%
2092       \string\newlabel{bib:end}{\arabic{page}}{page.\arabic{page}}}%
2093   \def\coppe@bibBegin{%
2094     \immediate\write\@auxout{%
2095       \string\newlabel{bib:begin}{\arabic{page}}{page.\arabic{page}}}%
2096   \def\coppe@mainBegin{%
2097     \immediate\write\@auxout{%
2098       \string\newlabel{front:pageno}{\Roman{page}}{page.\roman{page}}}%
2099   \def\coppe@hasLof{%
2100     \immediate\write\@auxout{%
2101       \string\newlabel{cat:lofflag}{:~il.;}{page.\roman{page}}}%
2102   }{%
2103     \def\coppe@bibEnd{%
2104       \immediate\write\@auxout{%
2105         \string\newlabel{bib:end}{\arabic{page}}}%
2106     \def\coppe@bibBegin{%
2107       \immediate\write\@auxout{%
2108         \string\newlabel{bib:begin}{\arabic{page}}}%
2109     \def\coppe@mainBegin{%
2110       \immediate\write\@auxout{%
2111         \string\newlabel{front:pageno}{\Roman{page}}}%
2112     \def\coppe@hasLof{%
2113       \immediate\write\@auxout{%
2114         \string\newlabel{cat:lofflag}{:~il.;}}%
2115   }%
2116 }
2117 % The reference list is now produced by biblatex's \printbibliography (see the
2118 % bibliography setup after babel, above). The former thebibliography
2119 % redefinition and its \bibindent are no longer used and have been removed; the
2120 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2121 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

2122 \newlength{\recuolongquote}%
2123 \setlength{\recuolongquote}{4cm}%
2124 \newenvironment*{longquote}[1][default]{%
2125   \list{}%
2126   \footnotesize%
2127   \addtolength{\leftskip}{\recuolongquote}%
2128   \item[]%
2129   \singlespacing%
2130   \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
2131 }{\endlist}%

2132 \newenvironment{theglossary}{%
2133   \if@twocolumn%
2134     \@restonecoltrue\onecolumn%
2135   \else%
2136     \@restonecolfalse%
2137   \fi%
2138   \@mkboth{\MakeUppercase\glossaryname}%
2139   {\MakeUppercase\glossaryname}%

```

```

2140 \chapter*{\glossaryname}%
2141 \addcontentsline{toc}{chapter}{\glossaryname}
2142 \list{}
2143 {\setlength{\listparindent}{0in}%
2144 \setlength{\labelwidth}{1.0in}%
2145 \setlength{\leftmargin}{1.5in}%
2146 \setlength{\labelsep}{0.5in}%
2147 \setlength{\itemindent}{0in}}%
2148 \sloppy}%
2149 {\if@restonecol\twocolumn\fi%
2150 \endlist}
2151 %
2152 \renewenvironment{theindex}{%
2153 \if@twocolumn
2154 \@restonecolfalse
2155 \else
2156 \@restonecoltrue
2157 \fi
2158 \twocolumn[\@makeschapterhead{\indexname}]%
2159 \mkboth{\MakeUppercase\indexname}%
2160 {\MakeUppercase\indexname}%
2161 \thispagestyle{plain}\parindent\z@
2162 \addcontentsline{toc}{chapter}{\indexname}
2163 \parskip\z@ \@plus .3\p@\relax
2164 \columnseprule \z@
2165 \columnsep 35\p@
2166 \let\item\@idxitem}
2167 {\if@restonecol\onecolumn\else\clearpage\fi}
2168 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2169 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2170 \newcommand\glossaryname{\coppemainstring{glossary}}
2171 %
2172 \newcommand\local@advisorstring{Orientador}
2173 \newcommand\foreign@advisorstring{Advisor}
2174 \ifthenelse{\boolean{maledoc}}{%
2175 \newcommand\local@approvedname{Examinado por}%
2176 }{%
2177 \newcommand\local@approvedname{Examinada por}%
2178 }
2179 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2180 \newcommand\local@deptstring{Programa}
2181 \newcommand\foreign@deptstring{Department}
2182 \newcommand\local@cityname{Rio de Janeiro}
2183 \newcommand\local@statename{RJ}
2184 \newcommand\local@countryname{Brasil}
2185 %
2186 \newcommand\frontcover@maintext{
2187 \sloppy\nohyphens{\local@doctype\ de \@degreename\
2188 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2189 ao Programa de P{\` o}s-gradua{\c c}{\~ a}o em \local@deptname,
2190 COPPE, da \local@universityname, como parte dos requisitos
2191 necess{\` a}rios {\` a} obten{\c c}{\~ a}o do t{\` i}tulo de
2192 \local@degname\ em \local@deptname.}
2193 }

```

```

2194 %
2195 \newcommand\frontpage@maintext{
2196   \noindent {\MakeUppercase\local@doctype}
2197   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2198   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2199   DE P{\` O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2200   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2201   NECESS{\` A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2202   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2203   {\MakeUppercase\local@deptname.\par}}%
2204 }
2205 %
2206 \newcommand\frontpage@bottomtext{%
2207   \begin{center}
2208     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2209     {\MakeUppercase\local@monthname\ DE \number\year}
2210   \end{center}%
2211 }
2212 %
2213 \newcommand\abstract@toptext{%
2214   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2215   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2216   \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
2217   necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
2218   \local@degname\ em Ci{\^ e}ncias (\@degree)}
2219 }
2220 \newcommand\foreignabstract@toptext{%
2221   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2222   COPPE/UFRJ as a partial fulfillment of the requirements for the
2223   degree of \foreign@degname\ of Science (\@degree)}
2224 }
2225 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

2226 \RequirePackage{setspace}
2227 % Apply the default line spacing now that setspace is available. The flag was

```

```

2228 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2229 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2230 \RequirePackage{hyperref}
2231 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
2232 % former natbib+bibtex setup. The numbers class option selects the numeric
2233 % citation style; otherwise the author-date default is used. natbib=true keeps
2234 % \citet/\citep available for users migrating from natbib.
2235 \if@coppenumeric
2236   \RequirePackage[backend=biber,datamodel=coppe,%
2237     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
2238 \else
2239   \RequirePackage[backend=biber,datamodel=coppe,%
2240     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
2241 \fi
2242 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
2243 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
2244 % "Bibliografia" (babel's default \bibname). The title is rendered through
2245 % \coppestring so it follows the current main language in BOTH the chapter
2246 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
2247 % the .toc write, and babel's \bibname caption override was equally fragile).
2248 \defbibheading{bibliography}{%
2249   \chapter*{\coppestring{references}}%
2250   \addcontentsline{toc}{chapter}{\coppestring{references}}}
2251 % The catalog page range (bib:begin/bib:end) was written by the old
2252 % thebibliography redefinition, which biblatex's \printbibliography does not
2253 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
2254 % begin label uses that hook and the end label is emitted by wrapping
2255 % \printbibliography to write it just after the reference list.
2256 \AtBeginBibliography{\coppe@bibBegin}
2257 \let\coppe@orig@printbibliography\printbibliography
2258 \renewcommand\printbibliography[1][]{\%
2259   \coppe@orig@printbibliography[#1]%
2260   \coppe@bibEnd}
2261 </class>
2262 <*glossary>
2263 actual '='
2264 quote '!'
2265 level '>'
2266 %%% delim_0 " , p. "
2267 delim_0 "\\dotfill "
2268 lethead_flag 0
2269 headings_flag 0
2270 preamble
2271 "\n\begin{theglossary}\n \\makeatletter"
2272 postamble
2273 "\n \\end{theglossary}\n"
2274 </glossary>
2275 <*class>
2276 \newcommand{\annex}{
2277   \renewcommand{\appendixname}{\coppestring{annex}}%
2278   \appendix
2279   \@coppeannextrue
2280 }

```

2281 $\langle$ /class $\rangle$

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```
2282  $\langle$ *dbx $\rangle$ 
2283 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
2284 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2285 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2286 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2287 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2288 % performance) are already in biblatex's standard datamodel.
2289 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2290 % Custom fields.
2291 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
2292 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2293 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
2294 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
2295 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
2296 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
2297 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2298 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
2299 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
2300 \DeclareDatamodelEntryfields{course,coppedegree}
2301 \DeclareDatamodelEntryfields{game,videogame,boardgame,software,movie,tvshow,video}%
2302 {artist,platform,director,producer,version}
2303 \DeclareDatamodelEntryfields[map]{scale}
2304 \DeclareDatamodelEntryfields[article]{caderno}
2305 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2306  $\langle$ /dbx $\rangle$ 
2307  $\langle$ *bbx $\rangle$ 
2308 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
2309 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2310 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2311 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2312 % so load setspace here as well (idempotent).
2313 \RequirePackage{setspace}
2314 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2315 % and numeric modes, so we derive from the standard style, not authoryear.
2316 \RequireBibliographyStyle{standard}
2317
2318 % Note: standard style never dashes repeated authors, which already matches the
2319 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2320 \ExecuteBibliographyOptions{%
2321 maxbibnames=99, % ABNT 4.3.1.3: list all authors by default
2322 giveninits=false,
2323 uniquename=false,
2324 uniquelist=false,
2325 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2326 sorting=nyt,
2327 }
2328
2329 % Language strings (Disponível em / Acesso em / In, etc.)
```



```

2330 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2331 \DeclareLanguageMapping{english}{english-coppe}
2332
2333 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2334 \DeclareNameAlias{author}{family-given}
2335 \DeclareNameAlias{editor}{family-given}
2336 \DeclareNameAlias{translator}{family-given}
2337 \DeclareNameAlias{default}{family-given}
2338 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2339 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2340
2341 % Uppercase the surname (family + prefix) only inside the bibliography list;
2342 % citations keep normal caps (post-2020 rule).
2343 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2344 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2345 % field names; the month stays abbreviated via \mkbibmonth.
2346 \protected\def\coppe@abntdate#1#2#3{%
2347   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2348   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2349   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
2350 \AtBeginBibliography{%
2351   \singlespacing % entries single-spaced internally (2.4)
2352   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2353   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2354   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2355   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2356   \let\mkbibdateshort\coppe@abntdate
2357 }
2358
2359 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2360 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2361 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2362 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2363 % article AND the next word are uppercased; otherwise just the first word. The
2364 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2365 % sentinel so no spurious space is left before the following punctuation.
2366 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2367 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2368   \coppeUCifart{#1}
2369   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2370   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2371 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2372   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2373 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2374 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2375 \protected\def\coppeUCifart#1{%
2376   \begingroup
2377   \lowercase{\def\coppeUC@w{#1}}%
2378   \edef\coppeUC@w{\coppeUC@w,\coppeUC@w}%
2379   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
2380   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2381   \endgroup}
2382 \DeclareFieldFormat{title}{%
2383   \ifbool{test}{\ifnameundef{labelname}}{}}

```

```

2384     {#1}%
2385     {\mkbibbold{#1}}
2386 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2387 % format receives the RAW title text (the title format above only sees the
2388 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2389 % first word here.
2390 \DeclareFieldFormat{titlecase}{%
2391   \ifbool{test}{\ifnameundef{labelname}}{ }
2392   {\coppeUCtitle{#1}}%
2393   {#1}}
2394 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
2395 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2396 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2397 % the title of a whole document, so override those back to bold.
2398 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2399 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2400 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2401 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2402 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2403 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
2404 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
2405
2406 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2407 \DeclareFieldFormat{pagetotal}{#1\addnspace p\adddot}
2408 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnspace f\adddot}
2409 \DeclareFieldFormat{pages}{p\adddot\addnspace#1}
2410
2411 % --- Periodical volume / number prefixes ---
2412 \DeclareFieldFormat[article]{volume}{v\adddot\addnspace#1}
2413 \DeclareFieldFormat[article]{number}{n\adddot\addnspace#1}
2414
2415 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2416 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2417
2418 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2419 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
2420
2421 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
2422 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2423 % differ: the newspaper name (journaltitle) and city, then either
2424 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2425 % "p. <pages>, <date>" (page before date, when none),
2426 % per NBR 6023 §4.2.3.6. ---
2427 \DeclareBibliographyDriver{article}{%
2428   \usebibmacro{bibindex}%
2429   \usebibmacro{begentry}%
2430   \usebibmacro{author/translator+others}%
2431   \setunit{\labelnamepunct}\newblock
2432   \usebibmacro{title}%
2433   \newunit\newblock
2434   \printfield{journaltitle}%
2435   \setunit{\addcomma\space}\printlist{location}%
2436   \iffieldequalstr{entrysubtype}{newspaper}
2437     {\iffieldundef{caderno}

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2438     {\setunit{\addcomma\space}\printfield{pages}%
2439     \setunit{\addcomma\space}\printdate}%
2440     {\setunit{\addcomma\space}\printdate%
2441     \setunit{\adddot\space}\printfield{caderno}%
2442     \setunit{\addcomma\space}\printfield{pages}}}%
2443     {\setunit{\addcomma\space}\printfield{volume}%
2444     \setunit{\addcomma\space}\printfield{number}%
2445     \setunit{\addcomma\space}\printfield{pages}%
2446     \setunit{\addcomma\space}\printdate}%
2447 \newunit\newblock
2448 \printfield{doi}%
2449 \newunit\newblock
2450 \usebibmacro{url+urldate}%
2451 \usebibmacro{finentry}}
2452
2453 % --- incollection: ABNT puts the host organizer before the booktitle, as
2454 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2455 % p. X-Y." The organizer role string is "Org./Orgs." (see the .ltx). ---
2456 \newbibmacro*{coppe:bookeditor}{%
2457   \ifnameundef{editor}
2458   {}
2459   {\printnames{editor}%
2460    \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2461    \clearname{editor}%
2462    \setunit{\adddot\space}}}%
2463 \newbibmacro*{coppe:partdriver}{%
2464   \usebibmacro{bibindex}\usebibmacro{begentry}%
2465   \usebibmacro{author/translator+others}%
2466   \setunit{\labelnamepunct}\newblock
2467   \usebibmacro{title}%
2468   \newunit\newblock
2469   \usebibmacro{in:}%
2470   \usebibmacro{coppe:bookeditor}%
2471   \printfield{booktitle}%
2472   \newunit\newblock
2473   \printfield{edition}%
2474   \newunit\newblock
2475   \printlist{location}%
2476   \setunit*{\addcolon\space}\printlist{publisher}%
2477   \setunit*{\addcomma\space}\printdate%
2478   \newunit\newblock
2479   \printfield{pages}%
2480   \setunit{\addcomma\space}\printfield{chapter}%
2481   \newunit\newblock
2482   \usebibmacro{url+urldate}%
2483   \usebibmacro{finentry}}
2484 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2485 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2486
2487 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2488 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2489
2490 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2491 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}

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```

2492 \DeclareBibliographyDriver{thesis}{%
2493   \usebibmacro{bibindex}\usebibmacro{begentry}%
2494   \usebibmacro{author}%
2495   \setunit{\labelnamepunct}\newblock
2496   \usebibmacro{title}%
2497   \newunit\newblock
2498   \printfield{type}\setunit*{\addspace}\printfield{course}%
2499   \newunit\newblock
2500   \printlist{institution}%
2501   \setunit*{\addcomma\space}\printlist{location}%
2502   \setunit*{\addcomma\space}\printdate%
2503   \newunit\newblock
2504   \printfield{pagetotal}%
2505   \newunit\newblock
2506   \printfield{note}%
2507   \newunit\newblock
2508   \usebibmacro{url+urldate}%
2509   \usebibmacro{finentry}}
2510 % --- Games / audiovisual: render the custom fields (artist, platform,
2511 %       director, producer, version) the generic misc driver ignores. ---
2512 \newbibmacro*{coppe:media}{%
2513   \iflistundef{artist}{ }\setunit{\addperiod\space}%
2514   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
2515   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
2516   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}%
2517   \iflistundef{director}{ }\setunit{\addperiod\space}%
2518   \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
2519   \iflistundef{producer}{ }\setunit{\addperiod\space}%
2520   \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
2521 \newbibmacro*{coppe:mediadriver}{%
2522   \usebibmacro{bibindex}\usebibmacro{begentry}%
2523   \usebibmacro{author/translator+others}%
2524   \setunit{\labelnamepunct}\newblock
2525   \usebibmacro{title}%
2526   \newunit\newblock
2527   \usebibmacro{coppe:media}%
2528   \newunit\newblock
2529   \printlist{publisher}%
2530   \setunit*{\addcomma\space}\printlist{location}%
2531   \setunit*{\addcomma\space}\printdate%
2532   \newunit\newblock
2533   \printfield{version}%
2534   \newunit\newblock
2535   \printfield{note}%
2536   \newunit\newblock
2537   \usebibmacro{url+urldate}%
2538   \usebibmacro{finentry}}
2539 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
2540 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
2541 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
2542 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
2543 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
2544 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2545 % imprenta) plus the mandatory scale ("Escala 1:N").

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2546 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
2547 \DeclareBibliographyDriver{map}{%
2548   \usebibmacro{bibindex}\usebibmacro{begentry}%
2549   \usebibmacro{author/translator+others}%
2550   \setunit{\labelnamepunct}\newblock
2551   \usebibmacro{title}%
2552   \newunit\newblock
2553   \printlist{location}%
2554   \setunit*{\addcolon\space}\printlist{publisher}%
2555   \setunit*{\addcomma\space}\printdate%
2556   \newunit\newblock
2557   \printfield{scale}%
2558   \newunit\newblock
2559   \usebibmacro{url+urldate}%
2560   \usebibmacro{finentry}}
2561 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2562 % number. date. The holder (depositante) is printed in normal case, not the
2563 % ALL-CAPS family treatment the bibliography gives the leading author.
2564 \newbibmacro*{coppe:depositor}{%
2565   \ifnameundef{holder}
2566   {}
2567   {\printtext{\bibstring{depositor}\addcolon\space}%
2568     \begingroup
2569       \def\mkbibnamefamily##1{##1}%
2570       \def\mkbibnameprefix##1{##1}%
2571       \printnames{holder}%
2572     \endgroup}}
2573 \DeclareBibliographyDriver{patent}{%
2574   \usebibmacro{bibindex}%
2575   \usebibmacro{begentry}%
2576   \usebibmacro{author/translator+others}%
2577   \setunit{\labelnamepunct}\newblock
2578   \usebibmacro{title}%
2579   \newunit\newblock
2580   \usebibmacro{coppe:depositor}%
2581   \newunit\newblock
2582   \printfield{type}%
2583   \setunit*{\addspace}\printfield{number}%
2584   \newunit\newblock
2585   \printdate%
2586   \newunit\newblock
2587   \usebibmacro{url+urldate}%
2588   \usebibmacro{finentry}}
2589 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2590 % give each a baseline (book-like for published docs: norma/partitura; misc for
2591 % the rest) so they print. Per-type ABNT refinements can follow later.
2592 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2593 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2594 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2595 % the law id + ementa (plain, not bold), then location and date.
2596 \newbibmacro*{coppe:legaldriver}{%
2597   \usebibmacro{bibindex}\usebibmacro{begentry}%
2598   \usebibmacro{author/translator+others}%
2599   \setunit{\labelnamepunct}\newblock

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2600 \usebibmacro{title}%
2601 \newunit\newblock
2602 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2603 \iffielddundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\newblock}{}
2604 \printlist{location}%
2605 \setunit{\addcomma\space}\printdate%
2606 \newunit\newblock
2607 \usebibmacro{url+urldate}%
2608 \usebibmacro{finentry}}
2609 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2610 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2611 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2612 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2613 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2614 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2615 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2616 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2617 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2618 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2619
2620 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2621 \DeclareSourcemap{%
2622   \maps[datatype=bibtex]{%
2623     % entry-type PT synonyms -> canonical English
2624     \map{step[typesource=livro,typetarget=book]}
2625     \map{step[typesource=artigo,typetarget=article]}
2626     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2627     \map{step[typesource=partedelivro,typetarget=incollection]}
2628     \map{step[typesource=emlivro,typetarget=inbook]}
2629     \map{step[typesource=verbete,typetarget=inreference]}
2630     \map{step[typesource=anais,typetarget=proceedings]}
2631     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
2632     \map{step[typesource=periodico,typetarget=periodical]}
2633     \map{step[typesource=relatorio,typetarget=report]}
2634     \map{step[typesource=relatoriotecnico,typetarget=report]}
2635     \map{step[typesource>manual,typetarget>manual]}
2636     \map{step[typesource=correspondencia,typetarget=letter]}
2637     \map{step[typesource=patente,typetarget=patent]}
2638     \map{step[typesource=diverso,typetarget=misc]}
2639     \map{step[typesource=jogo,typetarget=game]}
2640     \map{step[typesource=jogoeletronico,typetarget=videogame]}
2641     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
2642     \map{step[typesource=filme,typetarget=movie]}
2643     \map{step[typesource=serietv,typetarget=tvshow]}
2644     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
2645     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2646     % except map (declared in coppe.dbx).
2647     \map{step[typesource=legislacao,typetarget=legislation]}
2648     \map{step[typesource=atnormativo,typetarget=legislation]}
2649     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
2650     \map{step[typesource=documentojuridico,typetarget=legal]}
2651     \map{step[typesource=documentocivil,typetarget=legal]}
2652     \map{step[typesource=norma,typetarget=standard]}
2653     \map{step[typesource=partitura,typetarget=music]}

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2654 \map{\step[tourcesource=iconografico,typetarget=image]}
2655 \map{\step[tourcesource=obradearte,typetarget=artwork]}
2656 \map{\step[tourcesource=tridimensional,typetarget=artwork]}
2657 \map{\step[tourcesource=cartografico,typetarget=map]}
2658 \map{\step[tourcesource=mapa,typetarget=map]}
2659 \map{\step[tourcesource=audiovisual,typetarget=video]}
2660 \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
2661 \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
2662 \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
2663 % thesis family: retype to thesis and preset the (babel-aware) type word
2664 \map{\step[tourcesource=tese,typetarget=thesis]}
2665 \map{\step[tourcesource=dissertacao,typetarget=thesis]}
2666 \map{\step[tourcesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2667 \map{\step[tourcesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2668 \map{\step[tourcesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2669 \map{\step[tourcesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2670 \map{\step[tourcesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2671 \map{\step[tourcesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2672 \map{\step[tourcesource=teseph,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2673 \map{\step[tourcesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2674 % field PT synonyms (unaccented) -> canonical English
2675 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2676 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2677 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2678 \map{\step[fieldsource=titulo,fieldtarget=title]}
2679 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2680 \map{\step[fieldsource=autor,fieldtarget=author]}
2681 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2682 \map{\step[fieldsource=local,fieldtarget=location]}
2683 \map{\step[fieldsource=ano,fieldtarget=year]}
2684 \map{\step[fieldsource=data,fieldtarget=date]}
2685 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2686 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2687 \map{\step[fieldsource=numero,fieldtarget=number]}
2688 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2689 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2690 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2691 \map{\step[fieldsource=tipo,fieldtarget=type]}
2692 \map{\step[fieldsource=nota,fieldtarget=note]}
2693 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2694 \map{\step[fieldsource=curso,fieldtarget=course]}
2695 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2696 \map{\step[fieldsource=artista,fieldtarget=artist]}
2697 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2698 \map{\step[fieldsource=diretor,fieldtarget=director]}
2699 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2700 \map{\step[fieldsource=versao,fieldtarget=version]}
2701 \map{\step[fieldsource=escala,fieldtarget=scale]}
2702 }%
2703 }
2704
2705 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2706 \setlength{\bibhang}{0pt}
2707 \setlength{\bibitemsep}{\baselineskip}

```

```

2708
2709 </bbx>
2710 <*cbx>
2711 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
2712 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2713 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2714 \RequireCitationStyle{authoryear-comp}
2715
2716 % --- NBR 10520:2023 author separator -----
2717 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2718 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2719 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2720 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2721 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2722 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2723 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2724 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2725
2726 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2727 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2728 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2729 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2730 % so it is never added to the bibliography.
2731 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2732 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2733 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2734 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2735 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2736 \DeclareRobustCommand{\citetapud}[4][]{%
2737   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2738 \DeclareRobustCommand{\citepapud}[4][]{%
2739   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2740     \citeauthor{#4}\addcomma\space\citeyear{#4}%
2741     \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}}
2742 </cbx>
2743 <*lbxbr>
2744 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
2745 \InheritBibliographyExtras{brazilian}
2746 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2747 \DeclareBibliographyStrings{%
2748   inherit      = {brazilian},
2749   availablefrom = {{Dispon'\{i\}vel em}{Dispon'\{i\}vel em}},
2750   urlseen      = {{Acesso em}{Acesso em}},
2751   in           = {{In}{In}},
2752   editor       = {{Organizador}{Org\adddot}},
2753   editors      = {{Organizadores}{Orgs\adddot}},
2754   depositor    = {{Depositante}{Depositante}},
2755   coppescale   = {{Escala}{Escala}},
2756   mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2757   dscthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2758   phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2759   tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2760 }
2761 </lbxbr>

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```

2762 <*lboxen>
2763 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
2764 \InheritBibliographyExtras{english}
2765 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
2766 \DeclareBibliographyStrings{%
2767   inherit      = {english},
2768   availablefrom = {{Available from}{Available from}},
2769   urlseen      = {{Access on}{Access on}},
2770   in           = {{In}{In}},
2771   depositor    = {{Depositor}{Depositor}},
2772   coppescale   = {{Scale}{Scale}},
2773   mscdiss      = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
2774   dscthesi s   = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2775   phdthesis    = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2776   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
2777 }
2778 </lboxen>
2779 <*lboxes>
2780 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
2781 \InheritBibliographyExtras{spanish}
2782 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
2783 \DeclareBibliographyStrings{%
2784   inherit      = {spanish},
2785   availablefrom = {{Disponible en}{Disponible en}},
2786   urlseen      = {{Consultado el}{Consultado el}},
2787   in           = {{En}{En}},
2788   editor       = {{Editor}{Ed\addot}},
2789   editors      = {{Editores}{Eds\addot}},
2790   depositor    = {{Solicitante}{Solicitante}},
2791   coppescale   = {{Escala}{Escala}},
2792   mscdiss      = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
2793   dscthesi s   = {{Tesis Doctoral}{Tesis Doctoral}},
2794   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
2795   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
2796 }
2797 </lboxes>
2798 <*lboxfr>
2799 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
2800 \InheritBibliographyExtras{french}
2801 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
2802 \DeclareBibliographyStrings{%
2803   inherit      = {french},
2804   availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
2805   urlseen      = {{Consult\{e\} le}{Consult\{e\} le}},
2806   in           = {{Dans}{Dans}},
2807   editor       = {{\{\'Editeur}{\'Ed\addot}},
2808   editors      = {{\{\'Editeurs}{\'Eds\addot}},
2809   depositor    = {{D\'eposant}{D\'eposant}},
2810   coppescale   = {{\{\'Echelle}{\'Echelle}},
2811   mscdiss      = {{M\{\'emoire de Master}{M\{\'emoire de Master}},
2812   dscthesi s   = {{Th\{\'ese de Doctorat}{Th\{\'ese de Doctorat}},
2813   phdthesis    = {{Th\{\'ese de Doctorat}{Th\{\'ese de Doctorat}},
2814   tcc          = {{M\{\'emoire de fin d\'{\'etudes}{M\{\'emoire de fin d\'{\'etudes}},
2815 }

```

```

2816 </lbxfr>
2817 <*lbxit>
2818 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
2819 \InheritBibliographyExtras{italian}
2820 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2821 \DeclareBibliographyStrings{%
2822   inherit      = {italian},
2823   availablefrom = {{Disponibile da}{Disponibile da}},
2824   urlseen      = {{Consultato il}{Consultato il}},
2825   in           = {{In}{In}},
2826   editor       = {{Curatore}{Cur\adddot}},
2827   editors      = {{Curatori}{Curr\adddot}},
2828   depositor    = {{Depositante}{Depositante}},
2829   coppescale   = {{Scala}{Scala}},
2830   mscdiss      = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
2831   dscthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2832   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2833   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
2834 }
2835 </lbxit>
2836 <*langes>
2837 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
2838 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
2839 %% spanish slot) is active. Populates the dispatcher table for the
2840 %% spanish language and registers the biblatex language mapping.
2841 %% Institutional names (universityname/cityname/statename/countryname)
2842 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
2843 %% rosto are a Brazilian institutional template that does not translate.
2844 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
2845 \copperdefstring{spanish}{appendix}      {Ap\`endice}
2846 \copperdefstring{spanish}{annex}         {Anexo}
2847 \copperdefstring{spanish}{frame}         {Cuadro}
2848 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
2849 \copperdefstring{spanish}{source}        {Fuente}
2850 \copperdefstring{spanish}{program}       {Programa}
2851 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
2852 \copperdefstring{spanish}{references}    {Referencias}
2853 \copperdefstring{spanish}{in:}           {en}
2854 \copperdefstring{spanish}{art}           {Arte}
2855 \copperdefstring{spanish}{platform}      {Plataforma}
2856 \copperdefstring{spanish}{directedby}    {Direcci\`on}
2857 \copperdefstring{spanish}{producedby}    {Producci\`on}
2858 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
2859 \copperdefstring{spanish}{listsymbol}    {Lista de S\`imbolos}
2860 \copperdefstring{spanish}{glossary}      {Glosario}
2861 \copperdefstring{spanish}{advisor}       {Director}
2862 \copperdefstring{spanish}{advisors}      {Directores}
2863 \copperdefstring{spanish}{dept}          {Programa}
2864 \copperdefstring{spanish}{approvedmale}  {Examinado por}
2865 \copperdefstring{spanish}{approvedfemale}{Examinada por}
2866 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
2867 \copperdefstring{spanish}{cityname}      {Rio de Janeiro}
2868 \copperdefstring{spanish}{statename}     {RJ}
2869 \copperdefstring{spanish}{countryname}   {Brasil}

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2870 \copperdefstring{spanish}{monthname1} {enero}
2871 \copperdefstring{spanish}{monthname2} {febrero}
2872 \copperdefstring{spanish}{monthname3} {marzo}
2873 \copperdefstring{spanish}{monthname4} {abril}
2874 \copperdefstring{spanish}{monthname5} {mayo}
2875 \copperdefstring{spanish}{monthname6} {junio}
2876 \copperdefstring{spanish}{monthname7} {julio}
2877 \copperdefstring{spanish}{monthname8} {agosto}
2878 \copperdefstring{spanish}{monthname9} {septiembre}
2879 \copperdefstring{spanish}{monthname10}{octubre}
2880 \copperdefstring{spanish}{monthname11}{noviembre}
2881 \copperdefstring{spanish}{monthname12}{diciembre}
2882 \copperdefstring{spanish}{algorithm2eopt}{spanish}
2883 \copperdefstring{spanish}{babelname} {spanish}
2884 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
2885 %% language-mapping registration until \begin{document} so biblatex's
2886 %% \DeclareLanguageMapping is available.
2887 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
2888 \langes
2889 \*langfr
2890 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
2891 %% See coppe-lang-spanish.def for the structure. Institutional names
2892 %% remain Portuguese; only document-content strings translate.
2893 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
2894 \copperdefstring{french}{appendix} {Annexe}
2895 \copperdefstring{french}{annex} {Annexe}
2896 \copperdefstring{french}{frame} {Encadr'e}
2897 \copperdefstring{french}{listframe} {Liste des encadr'es}
2898 \copperdefstring{french}{source} {Source}
2899 \copperdefstring{french}{program} {Programme}
2900 \copperdefstring{french}{listprogram} {Liste des programmes}
2901 \copperdefstring{french}{references} {R'ef'erenes}
2902 \copperdefstring{french}{in:} {dans}
2903 \copperdefstring{french}{art} {Illustration}
2904 \copperdefstring{french}{platform} {Plateforme}
2905 \copperdefstring{french}{directedby} {R'ealis'e par}
2906 \copperdefstring{french}{producedby} {Produit par}
2907 \copperdefstring{french}{listabbreviation}{Liste des abr'eviations et acronymes}
2908 \copperdefstring{french}{listsymbol} {Liste des symboles}
2909 \copperdefstring{french}{glossary} {Glossaire}
2910 \copperdefstring{french}{advisor} {Directeur}
2911 \copperdefstring{french}{advisors} {Directeurs}
2912 \copperdefstring{french}{dept} {Programme}
2913 \copperdefstring{french}{approvedmale} {Examin'e par}
2914 \copperdefstring{french}{approvedfemale} {Examin'ee par}
2915 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
2916 \copperdefstring{french}{cityname} {Rio de Janeiro}
2917 \copperdefstring{french}{statename} {RJ}
2918 \copperdefstring{french}{countryname} {Brasil}
2919 \copperdefstring{french}{monthname1} {janvier}
2920 \copperdefstring{french}{monthname2} {f'evrier}
2921 \copperdefstring{french}{monthname3} {mars}
2922 \copperdefstring{french}{monthname4} {avril}
2923 \copperdefstring{french}{monthname5} {mai}

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2924 \copperdefstring{french}{monthname6} {juin}
2925 \copperdefstring{french}{monthname7} {juillet}
2926 \copperdefstring{french}{monthname8} {août}
2927 \copperdefstring{french}{monthname9} {septembre}
2928 \copperdefstring{french}{monthname10}{octobre}
2929 \copperdefstring{french}{monthname11}{novembre}
2930 \copperdefstring{french}{monthname12}{décembre}
2931 \copperdefstring{french}{algorithm2eopt}{french}
2932 \copperdefstring{french}{babelname} {french}
2933 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
2934 \end{langfr}
2935 \end{langit}
2936 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
2937 %% See coppe-lang-spanish.def for the structure. The algorithm2e
2938 %% language option for Italian is spelled "italiano" -- that's the
2939 %% spelling algorithm2e ships, even though babel calls the language
2940 %% "italian".
2941 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
2942 \copperdefstring{italian}{appendix} {Appendice}
2943 \copperdefstring{italian}{annex} {Allegato}
2944 \copperdefstring{italian}{frame} {Riquadro}
2945 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
2946 \copperdefstring{italian}{source} {Fonte}
2947 \copperdefstring{italian}{program} {Programma}
2948 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
2949 \copperdefstring{italian}{references} {Riferimenti}
2950 \copperdefstring{italian}{in:} {in}
2951 \copperdefstring{italian}{art} {Illustrazione}
2952 \copperdefstring{italian}{platform} {Piattaforma}
2953 \copperdefstring{italian}{directedby} {Regia di}
2954 \copperdefstring{italian}{producedby} {Prodotto da}
2955 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
2956 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
2957 \copperdefstring{italian}{glossary} {Glossario}
2958 \copperdefstring{italian}{advisor} {Relatore}
2959 \copperdefstring{italian}{advisors} {Relatori}
2960 \copperdefstring{italian}{dept} {Programma}
2961 \copperdefstring{italian}{approvedmale} {Esaminato da}
2962 \copperdefstring{italian}{approvedfemale} {Esaminata da}
2963 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
2964 \copperdefstring{italian}{cityname} {Rio de Janeiro}
2965 \copperdefstring{italian}{statename} {RJ}
2966 \copperdefstring{italian}{countryname} {Brasil}
2967 \copperdefstring{italian}{monthname1} {gennaio}
2968 \copperdefstring{italian}{monthname2} {febbraio}
2969 \copperdefstring{italian}{monthname3} {marzo}
2970 \copperdefstring{italian}{monthname4} {aprile}
2971 \copperdefstring{italian}{monthname5} {maggio}
2972 \copperdefstring{italian}{monthname6} {giugno}
2973 \copperdefstring{italian}{monthname7} {luglio}
2974 \copperdefstring{italian}{monthname8} {agosto}
2975 \copperdefstring{italian}{monthname9} {settembre}
2976 \copperdefstring{italian}{monthname10}{ottobre}
2977 \copperdefstring{italian}{monthname11}{novembre}

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2978 \copperdefstring{italian}{monthname12}{dicembre}
2979 \copperdefstring{italian}{algorithm2eopt}{italiano}
2980 \copperdefstring{italian}{babelname} {italian}
2981 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
2982 </langit>
2983 <*numbbx>
2984 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
2985 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
2986 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
2987 % the PT->EN source map and the language mappings) and only adds the [n] list
2988 % label and the numeric sorting on top.
2989 \RequireBibliographyStyle{coppe}
2990
2991 % ABNT numeric system lists references in the order they are first cited
2992 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
2993 % numbered list. 'labelnumber' generates the [n] field.
2994 \ExecuteBibliographyOptions{labelnumber,sorting=none}
2995
2996 % Bracketed labels, both in the list and for any shorthand entries.
2997 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
2998 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
2999
3000 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
3001 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3002 \defbibenvironment{bibliography}
3003 {
3004   \list
3005     {\printtext[labelnumberwidth]{%
3006       \printfield{labelprefix}%
3007       \printfield{labelnumber}}}
3008     {\setlength{\labelwidth}{\labelnumberwidth}%
3009       \setlength{\leftmargin}{\labelwidth}%
3010       \setlength{\labelsep}{\biblabelsep}%
3011       \addtolength{\leftmargin}{\labelsep}%
3012       \setlength{\itemsep}{\bibitemsep}%
3013       \setlength{\parsep}{\bibparsep}}%
3014     {\renewcommand*{\makelabel}[1]{\hss#1}}
3015   {\endlist}
3016   {\item}
3017 </numbbx>
3018 <*numcbx>
3019 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3020 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3021 % numbers [1], compressed and sorted within a single cite group. This is the
3022 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3023 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
3024 % \citet -> Author [1].
3025 \RequireCitationStyle{numeric-comp}
3026 </numcbx>

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Acknowledgments

Thanks to all COPPE_{TE}X users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Change History

v0.0	General: Creation Date.	1	v1.0	General: First COPPETeX release. .	1
v0.1	General: Documentation: bibliography fixed, title translation.	1	v2.0	General: COPPETeX release 2.0. . .	1
	Sourceforge submission.	1	v2.1	General: COPPETeX release 2.1: Matching the new rules.	1
v0.2	General: Unification of the code for the list of symbols and abbreviations.	1		\advisor: Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors.	37
v0.3	General: Added ‘draft’ option. . . .	1	v2.2	General: Matching new guidelines, including new logo.	1
	\maketitle: Added number of examiners test.	41	v2.2.2	General: Fixed some text constants in .bib and documented it here.	1
	Generalization.	41	v3.0	General: Added support for monographs in English.	36
v0.4	General: Beta documentation. . . .	1		New approval page layout. . . .	42
v0.5	abstract: Changed from macro to environment.	45		\department: Added new course on Nanotechnology.	37
	\backmatter: Added mainmatter pages counter.	39		\examiner: Examiners expansion without degree.	38
	foreignabstract: Changed from macro to environment.	46	v3.1	\department: Included a sort key	

	in symb1	37	v4.0	
v3.2	General: Fixed version problem between cls and dtx.	1		General: Class-wide string dispatcher (copperdefstring, coppestring, coppemainstring, coppeforeignstring) replaces scattered iflanguage and if english branches; pt/en outputs remain byte-identical. . .
v3.3	General: Extend abbrev to work like symb and accept a sorting key . .	1		1
v3.4	General: Some examples for figures, tables, longtables, etc. . .	1		Multilingual model: three fixed slots (main, foreign, optional third); five built-in language options (brazilian, english, spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.
v3.4.1	General: Fix english option to consider brazilian a second language, describe it in document	36		1
v3.5	General: New command annex . .	66		New user API: titlein, usecoppelanguage, brazilianabstract environment. . .
v3.5.1	General: New command annex now supports brazilian or english . .	66		1